



2025第十九届中国汽车产业论坛

The 19th China Automotive Industry Forum 2025

AI驱动的汽车产业数智化全球能力构筑：产品再造、组织进化与生态协同
Building Global Capabilities for the AI-Driven Digital Transformation of the Automotive Industry:
Redefining Products, Evolving Organizations, and Fostering Ecosystem Synergy

论坛指定用车



论坛支持



智力支持



媒体支持



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| 中欧校友人力资源
管理研究会(上海)

 CEIBS

| 中欧上海校友会
电子半导体协会

SDV-EV Industrial Competition and International Alliances

Capability-Architecture-Performance (CAP) Approach

Dec. 5 2025

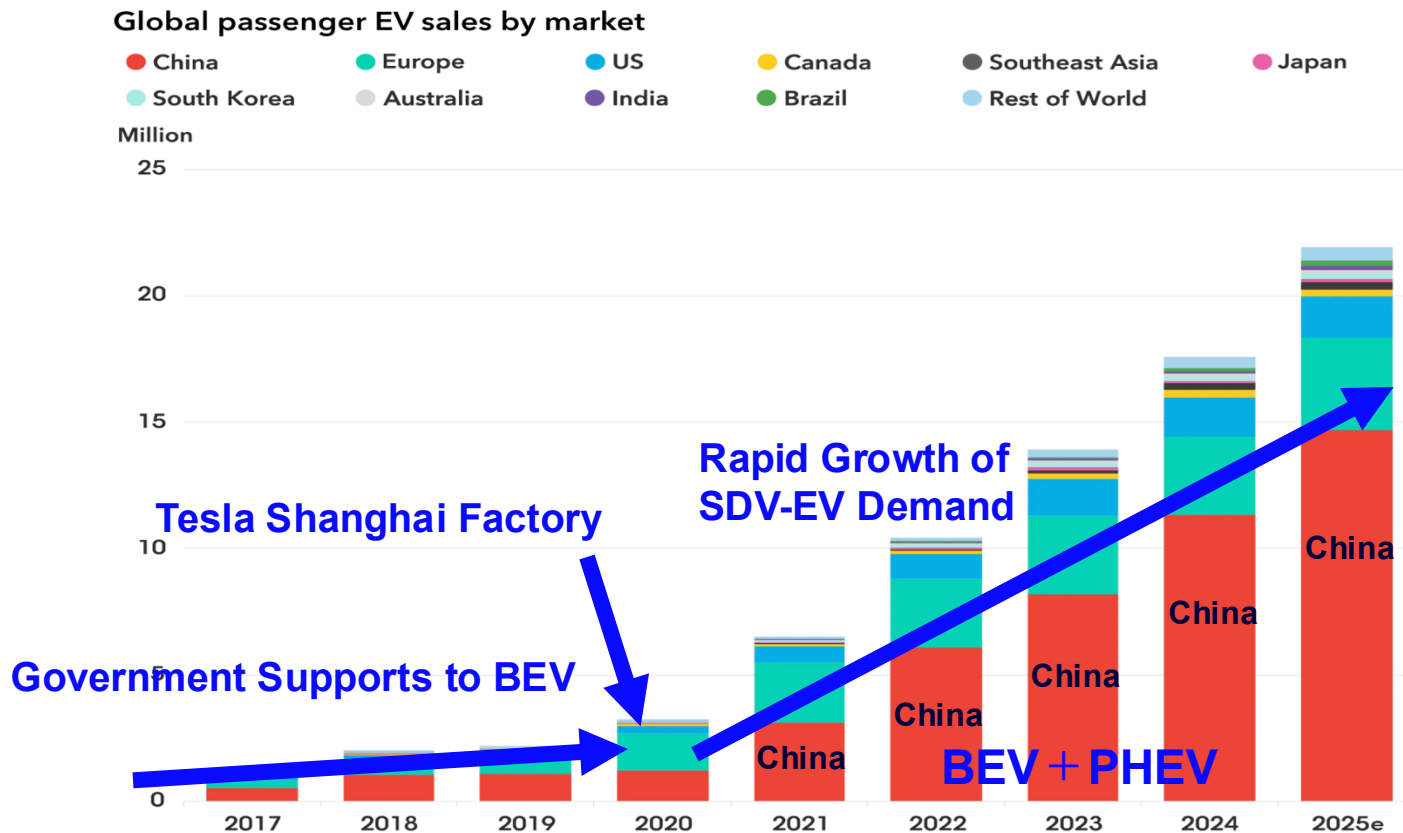
Takahiro Fujimoto

Professor, Waseda university,
Emeritus professor, Tokyo university

Pin (Peter) Zhou

Co-Chairman, Automotive Industry Alumni Association (AIAA)
Executive Vice President, CEIBS Alumni Auto Association (CAAA)

The industrial phenomenon that should be explained: Rapid expansion of China's BEV+PHEV market after 2020



Source: BloombergNEF, MarkLines, Jato Dynamics

Note: Includes battery electric and plug-in hybrid passenger vehicles. 2025 is BNEF's forecast for the year.

Questions (and Answer)

Q1 Why did German and Japanese automakers, with **design-based comparative advantage** in high-performance automobile hardware, lagged behind their US (e.g., Tesla) and Chinese rivals in developing **SDV-EV**?

(A1: *Design-based comparative disadvantage* in *modular* application software)

Q2 How could Chinese EV specialists quickly establish its **software-hardware integration** of **SDV-EV**?

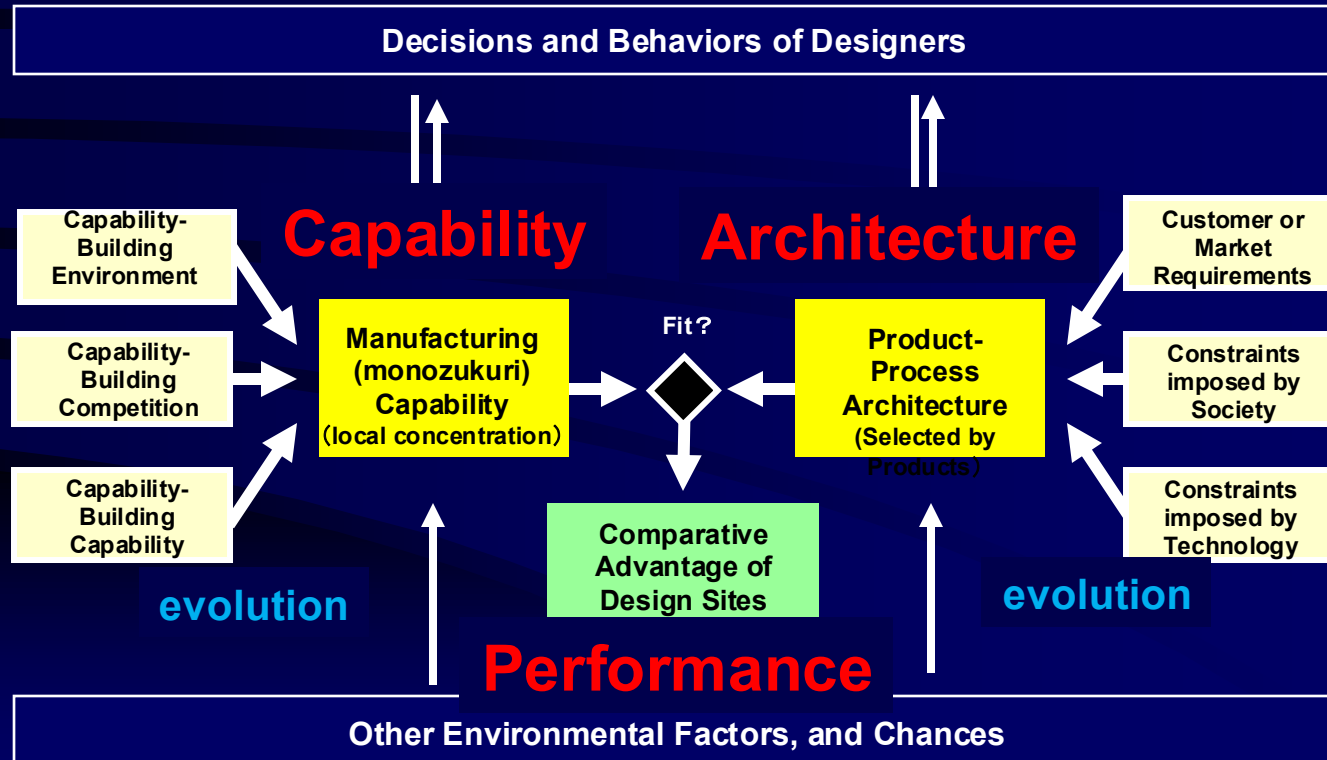
(A2: *Design-based comparative advantage* in *modular* application software)

Q3 How can German and Japanese automakers quickly catch up with the US and Chinese rivals for developing SDV with multiple powertrains (EV, HEV, ICE, etc.)?

(A3: *International inter-firm **alliance** for in-car & out-car software integration*)

Basic Framework for Analyzing Industry Performances:

Capability-Architecture-Performance (CAP) Approach



Design-based Comparative Advantage

CAP Approach to Industrial Performance

Balance between

product architecture (A)

and

organizational capability (C)

results in

competitive performance (P)

design-based comparative advantage

C: *Capabilities* Differ between Countries

History matters.

Labor market conditions during the country's high-growth era

US, China: Fluid labor market & abundant labor force in high-growth era

- Quick matching of skills and tasks → *Short-term employment*
- *Single-skilled workers*, standardization, simplification, specialization,
- *mass production*, Fordism → *Specialization capability*

Japan, Germany: Stable labor market & chronic labor shortage

- High switching cost of labor → *Long-term employment*
- *Multi-skilled workers, teamwork*, multi-tasking, coordination,
- flexible production, *Toyota/Lean System* → *Coordination capability*

A: Architectures Differ between Products

Functional requirements and constraints matter

Tougher requirements/constraints → *complex products* with *integral architecture*

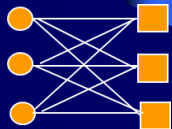
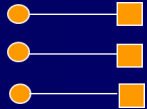
Modular Hardware/software: Milder functional requirements & constraints

Weight-free (nearly) → Physical laws do NOT dictate product design

- Combinations of *language and logic* as design rules
- Structural/functional *independence* between parts & design elements
- Relatively *open-modular architecture* (*coordination-saving* design)

Integral Hardware/software: Stricter functional requirements & constraints

- *Weighty physical* artifacts → *Physical laws* dictate product design
- Complex *structure-function causality* between physical components
- Structural/functional *interdependence* between parts & design elements
- Relatively *closed-integral architecture* (*coordination-intensive* design)



A: Architectures Differ between Products

Yin-Yang
analogy

Functional requirements and constraints matter

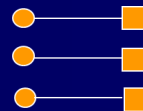
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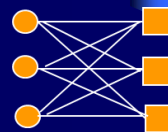
陽 Yang



Integral Hardware/software: Stricter functional requirements & constraints

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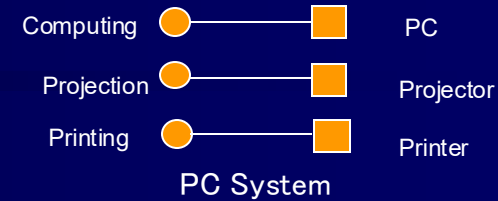
陰 Yin



A: Basic Types of Product-Process Architectures

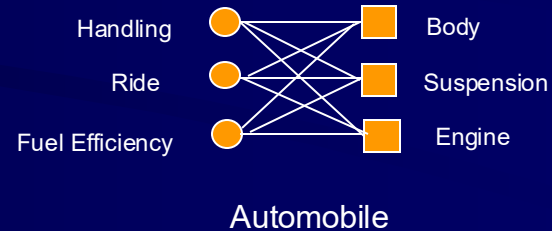
Modular architecture

one-to-one correspondence
between functional
and structural elements



Integral architecture

many-to-many correspondence
between the functional
and structural elements



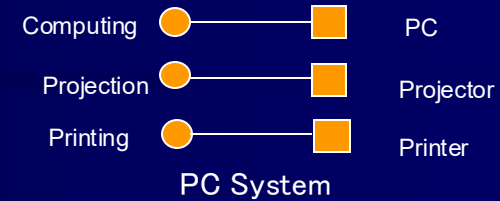
Open architecture: connected by *industry-standard open interfaces*

Closed architecture: Connected by firm-specific *closed interfaces*

A: Basic Types of Product-Process Architectures

Modular architecture

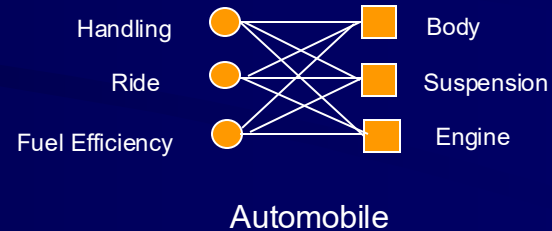
one-to-one correspondence
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陽 Yang

Integral architecture

many-to-many correspondence
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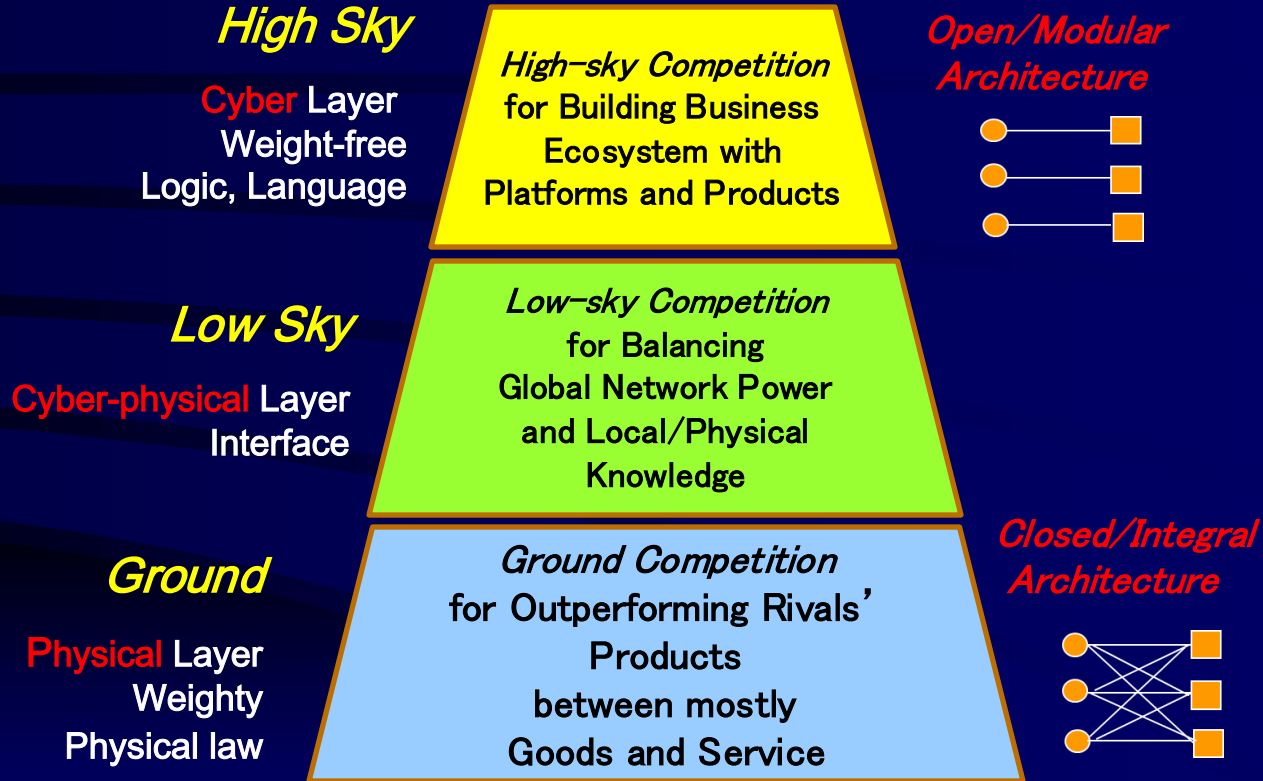


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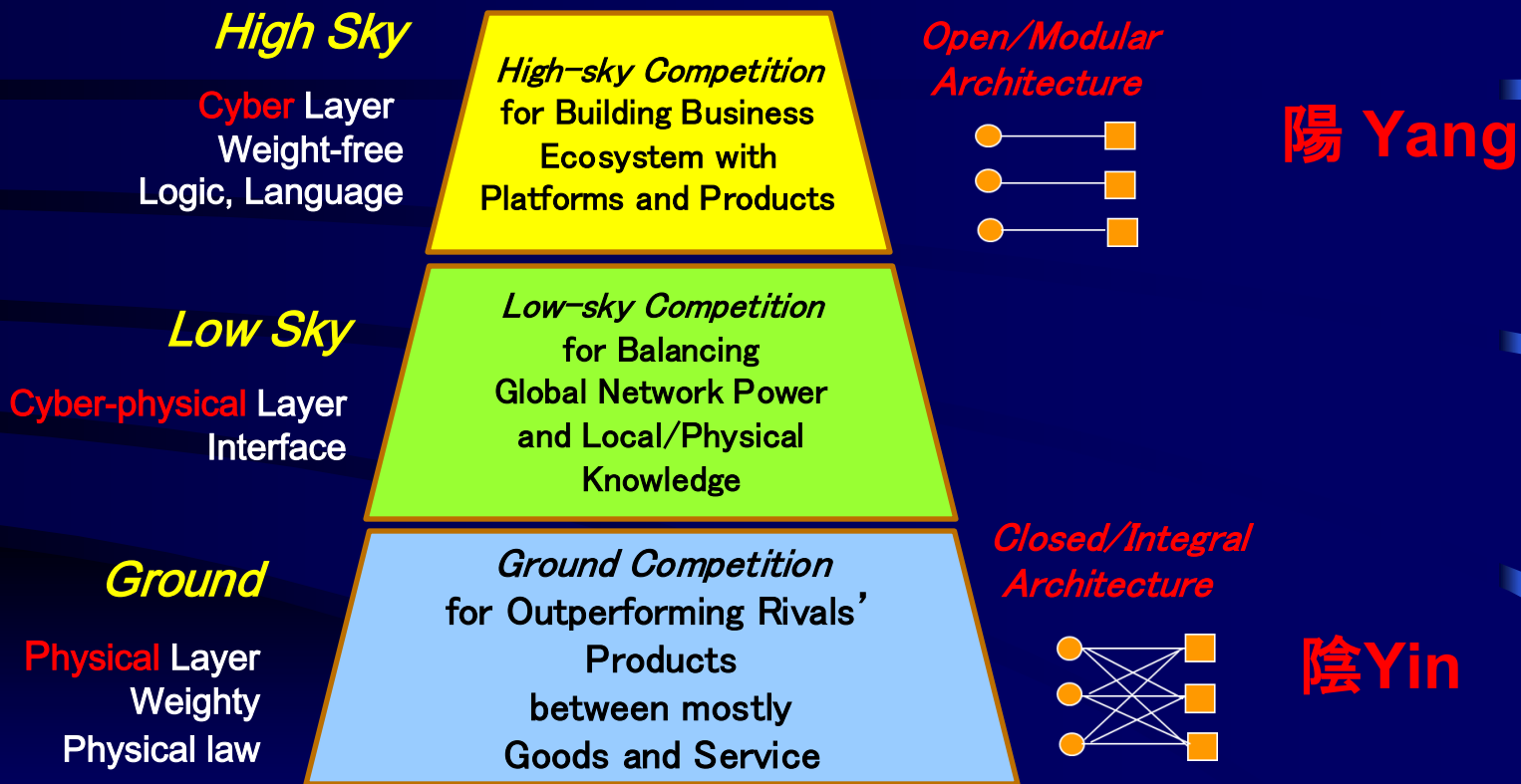
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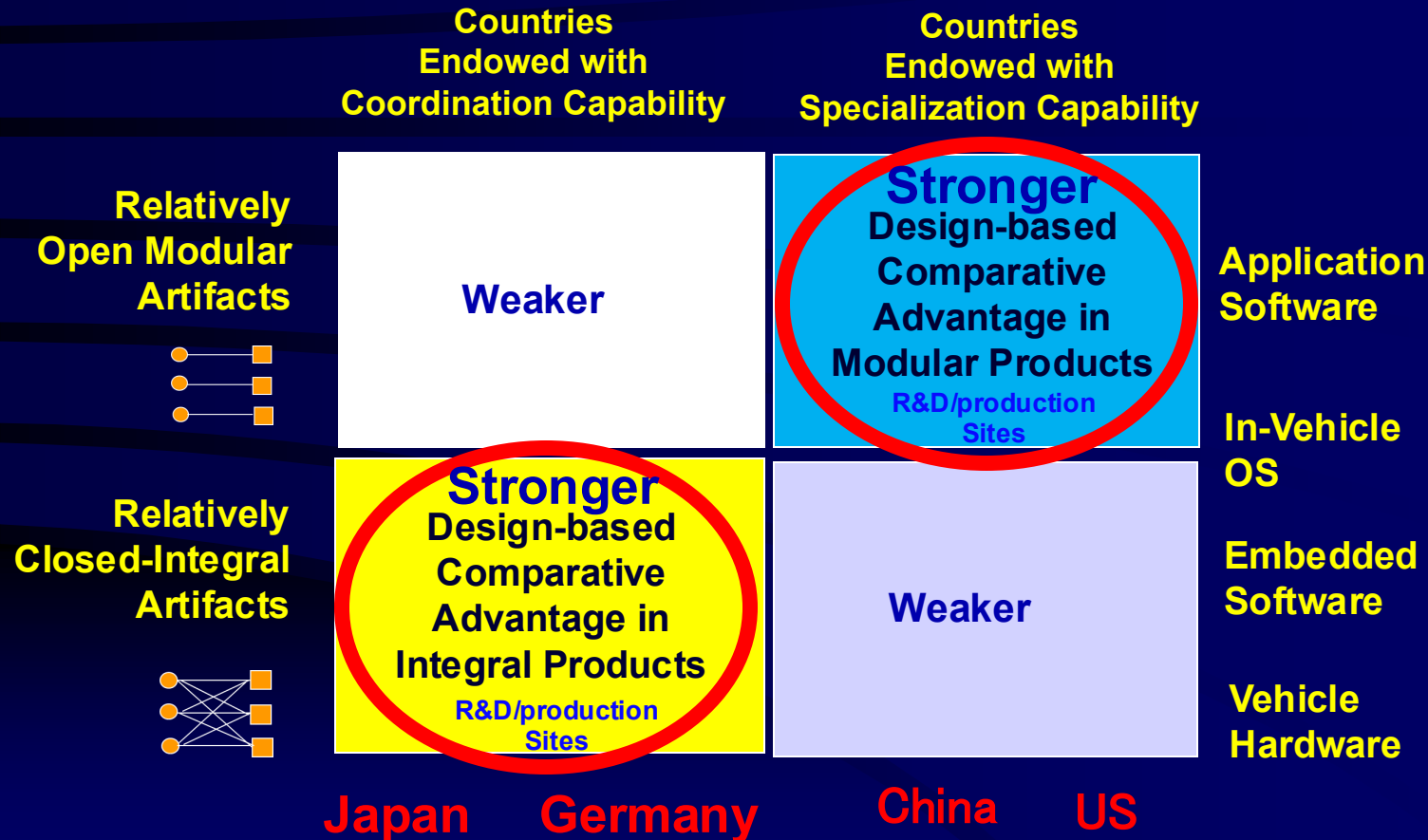
A: Design Layers and Architectures : High Sky, Low Sky, Ground



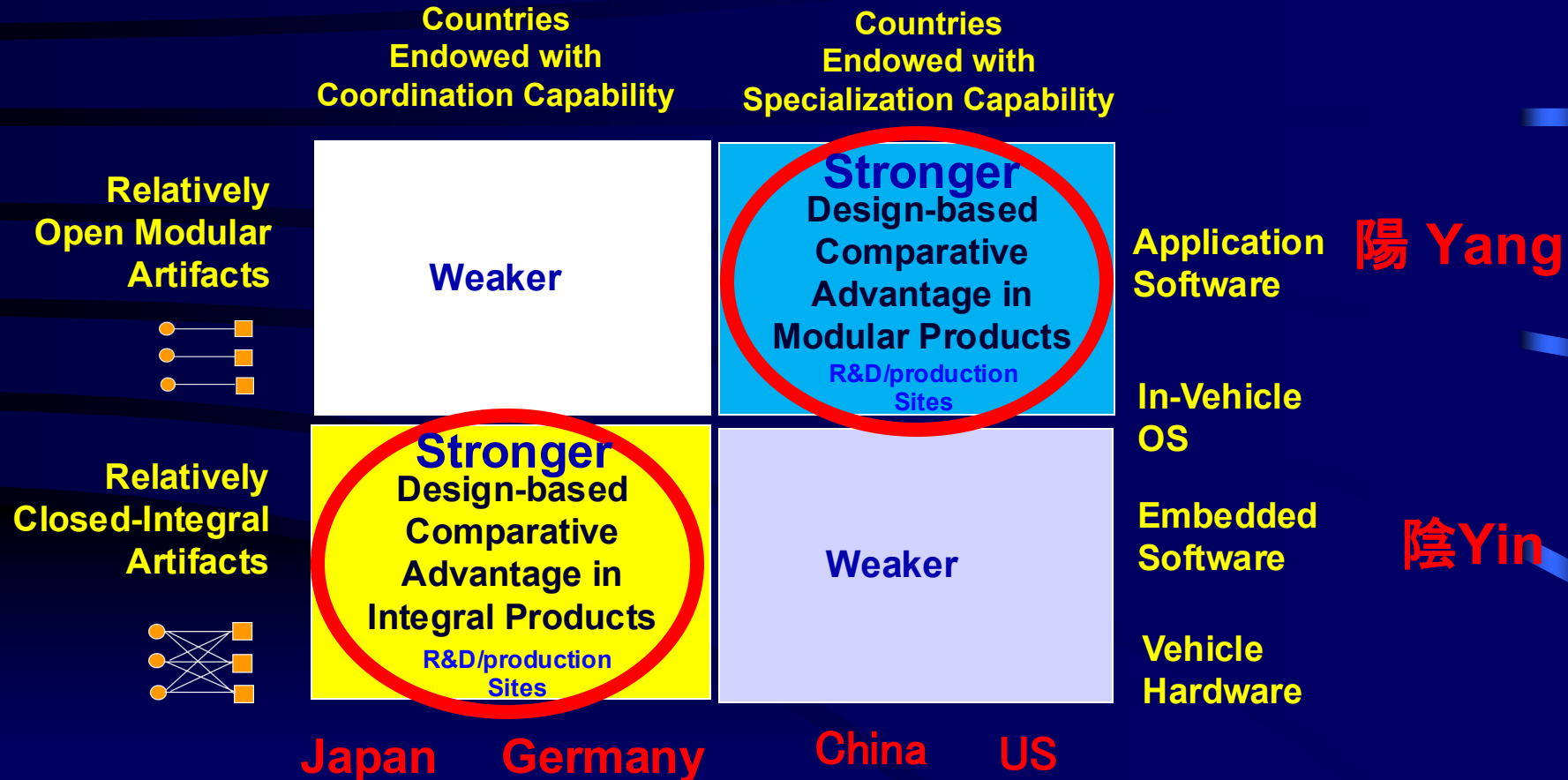
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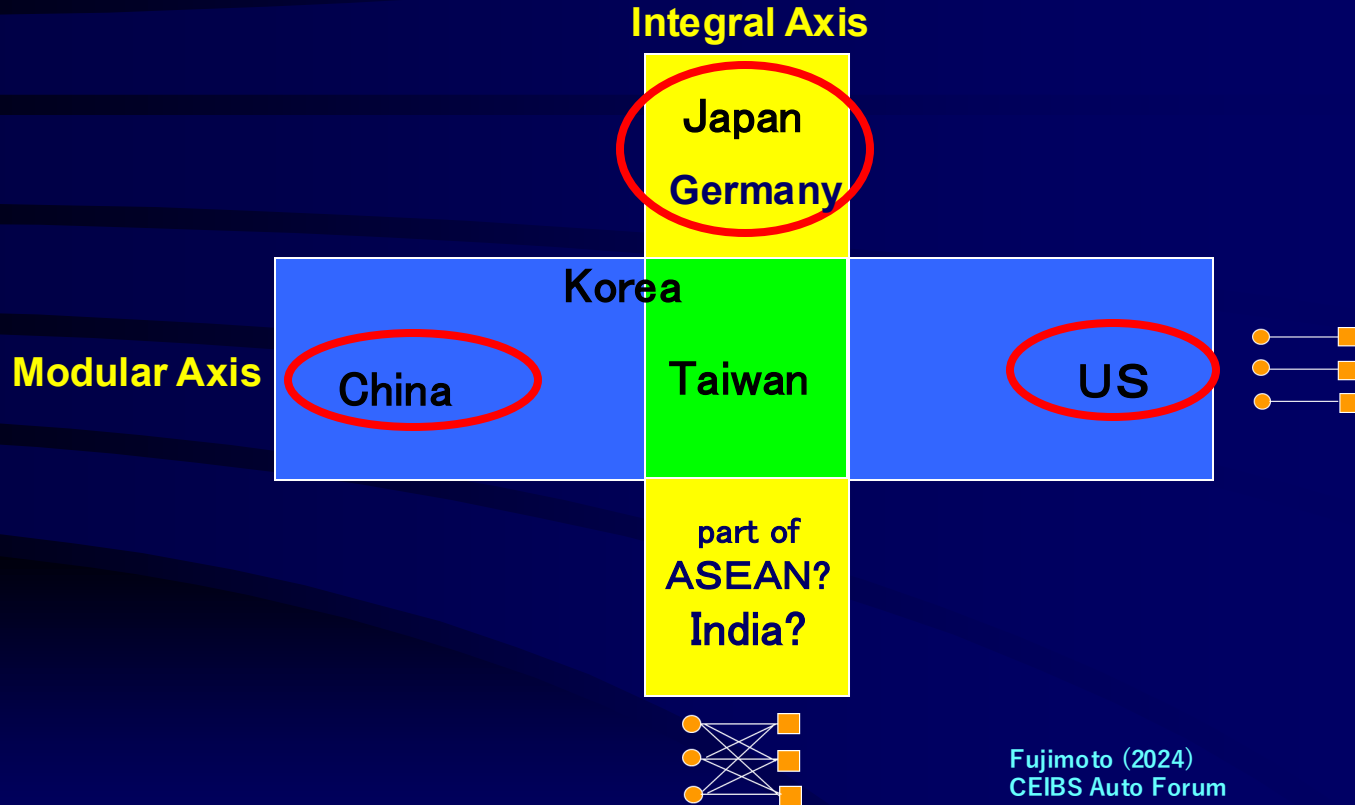
P: Design-based Comparative Advantage in Automobile Hardware & Software



P: Design-based Comparative Advantage in Automobile Hardware & Software



P: Performance Geopolitics *Design-based Comparative Advantage Predicted*

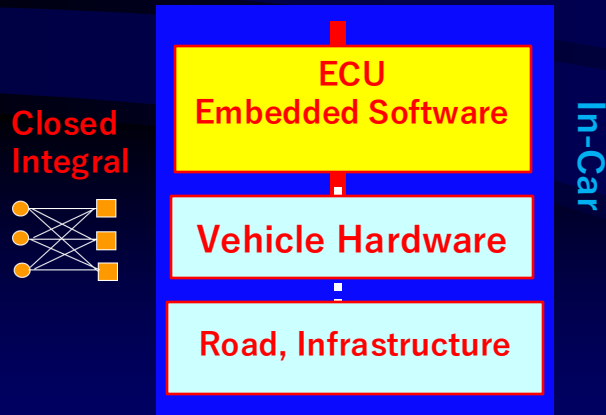


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CEIBS Auto Forum

Design Layers of Traditional Cars and SDVs

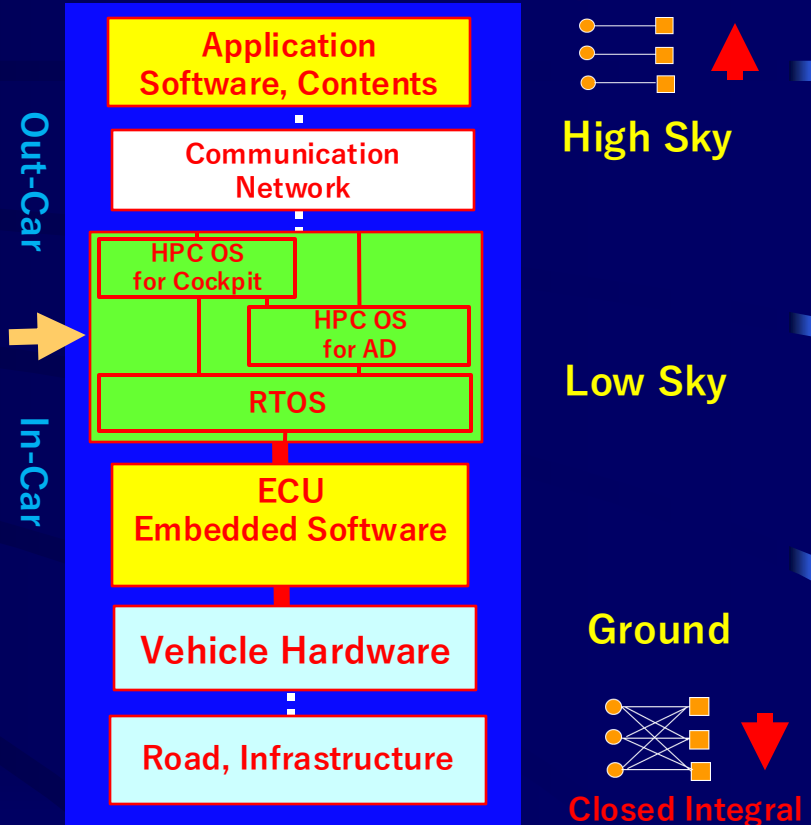
— closed interface
..... open interface

① Traditional Cars' Design Layers (Closed)



Vehicle OS →

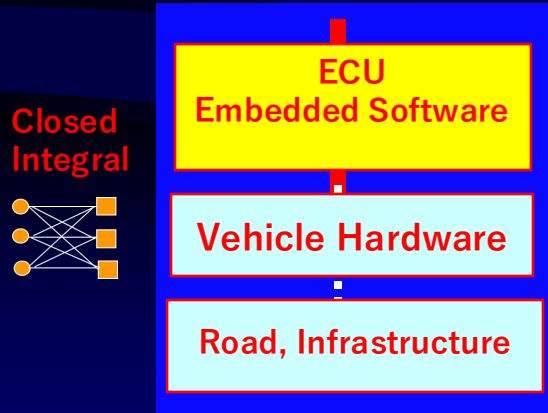
② SDVs' Design Layers (Closed and Open)



Design Layers of Traditional Cars and SDVs

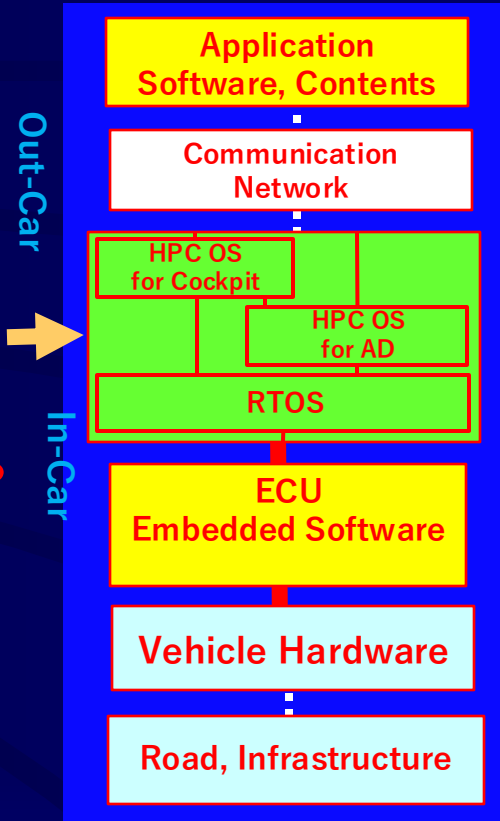
—— closed interface
..... open interface

① Traditional Cars' Design Layers (Closed)



Vehicle OS →
Yin-Yang Integration?
(陰陽融合)

② SDVs' Design Layers (Closed and Open)



Open Modular

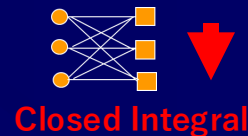


High Sky

Low Sky

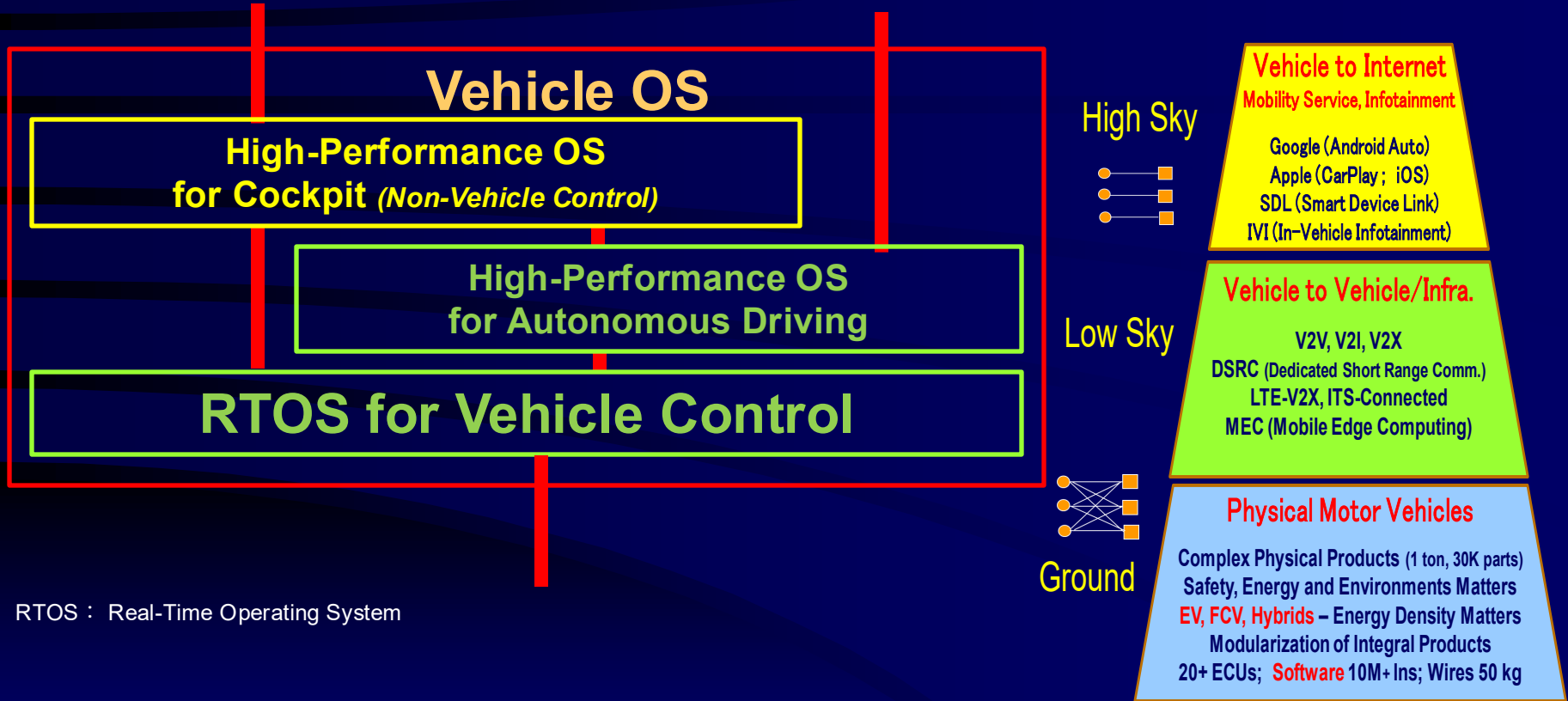
陰Yin

Ground



Closed Integral

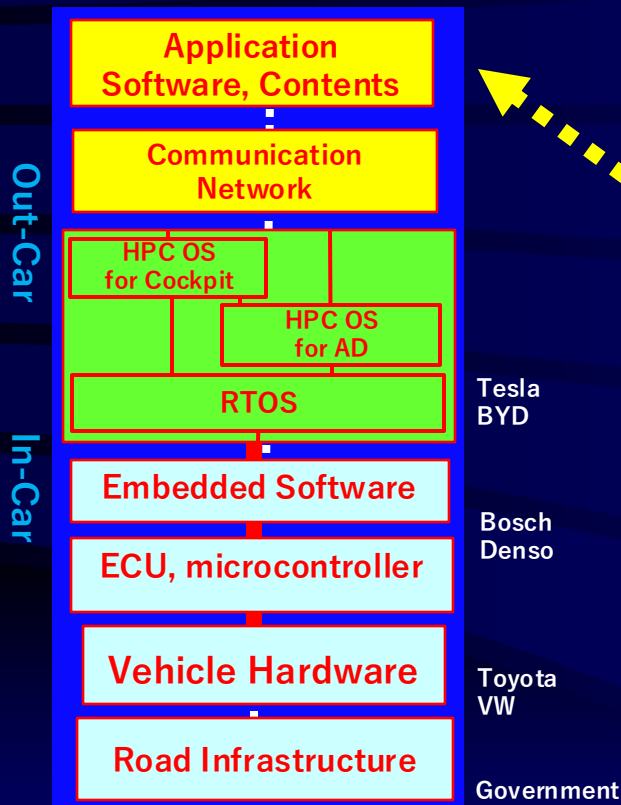
Three Layers of Vehicle OS



RTOS : Real-Time Operating System

Integrating SDV Layers and AI Layers

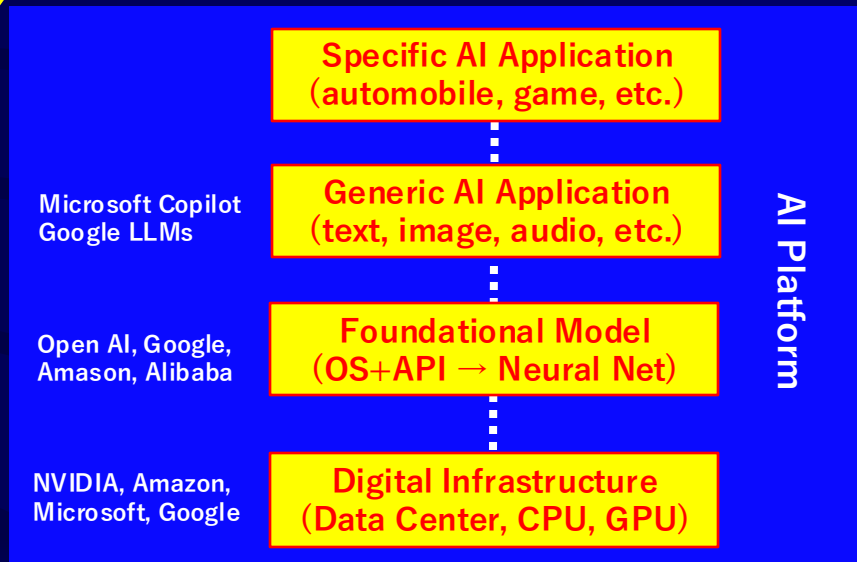
SDVs' Design Layers



Fujimoto (2024)
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— closed interface
..... open interface

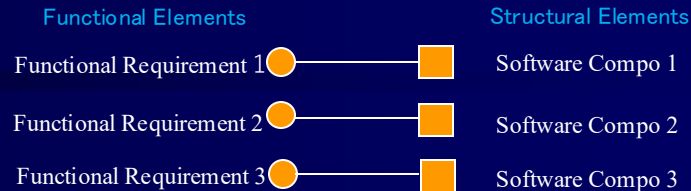
AI Platform's Design Layers (flip over!)



Cusumano (2023) Generative AI as a New Innovation Platform
Communications of the ACM. October 2023

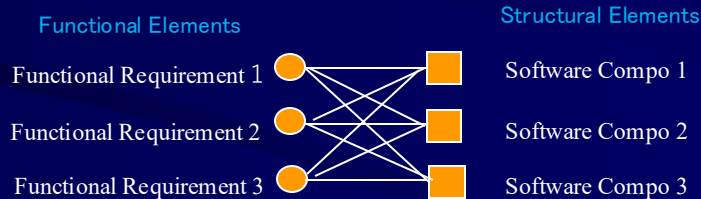
Software's Architectural Types

Software with *Modular Architecture*



e.g., **AI, OS, Applications**, etc.

Software with *Integral Architecture*

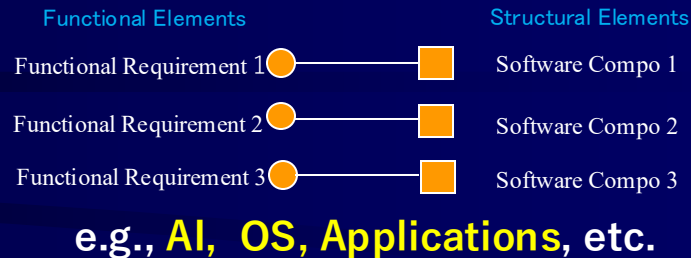


e.g., **Embedded Software in ECU**
for High Performance Vehicles

Based on Ulrich 1995

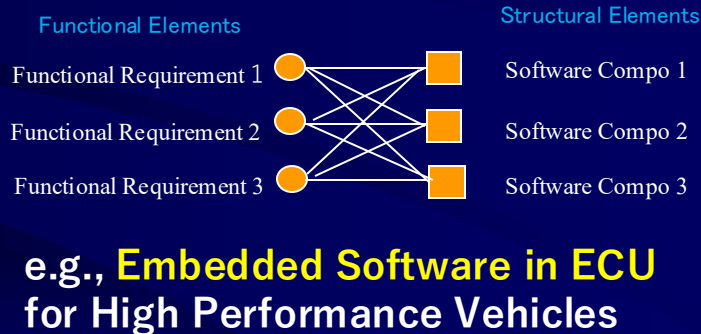
Software's Architectural Types

Software with *Modular Architecture*



陽 Yang

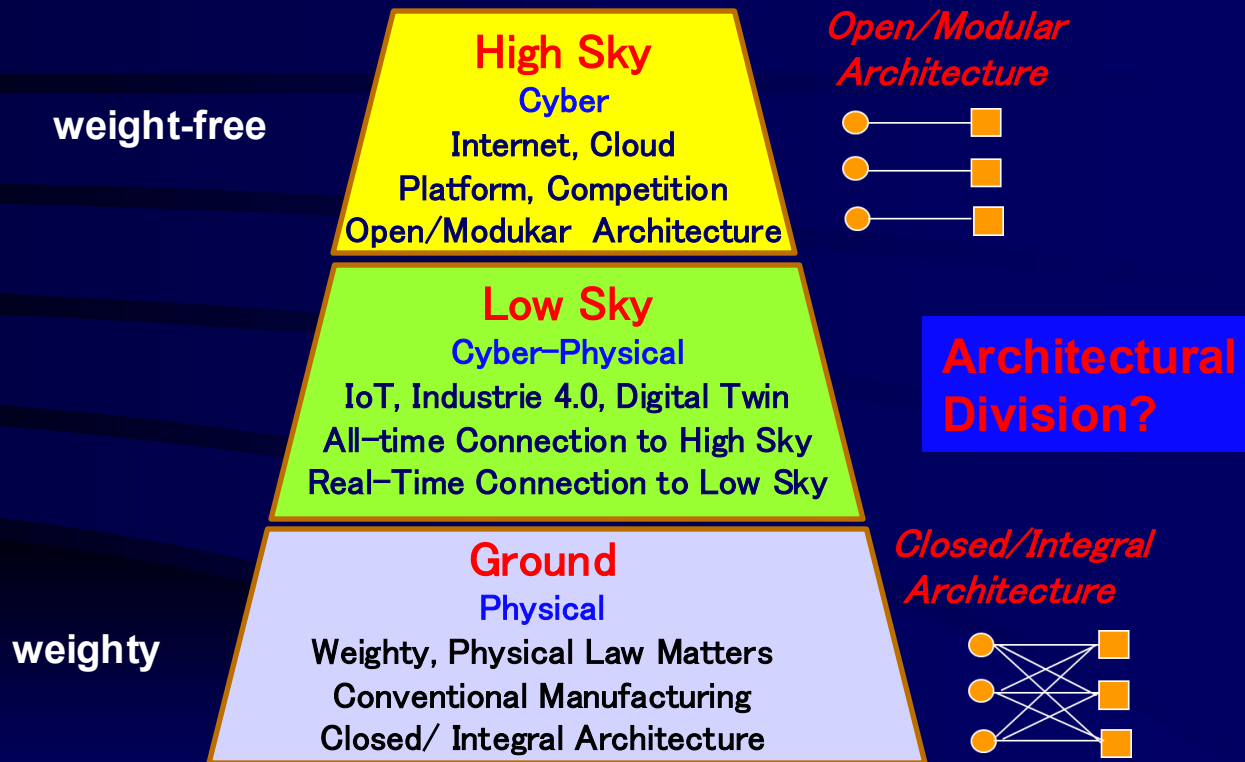
Software with *Integral Architecture*



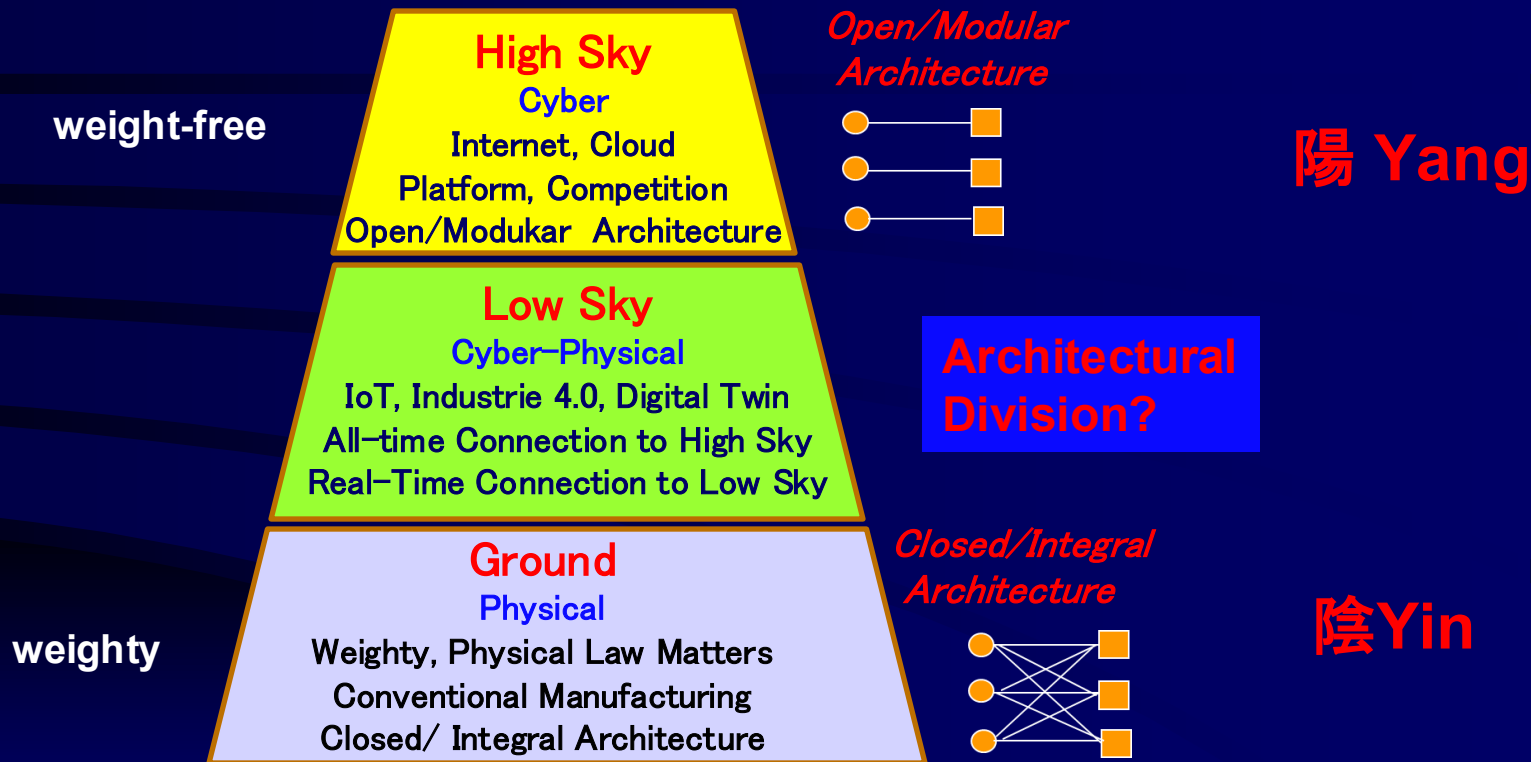
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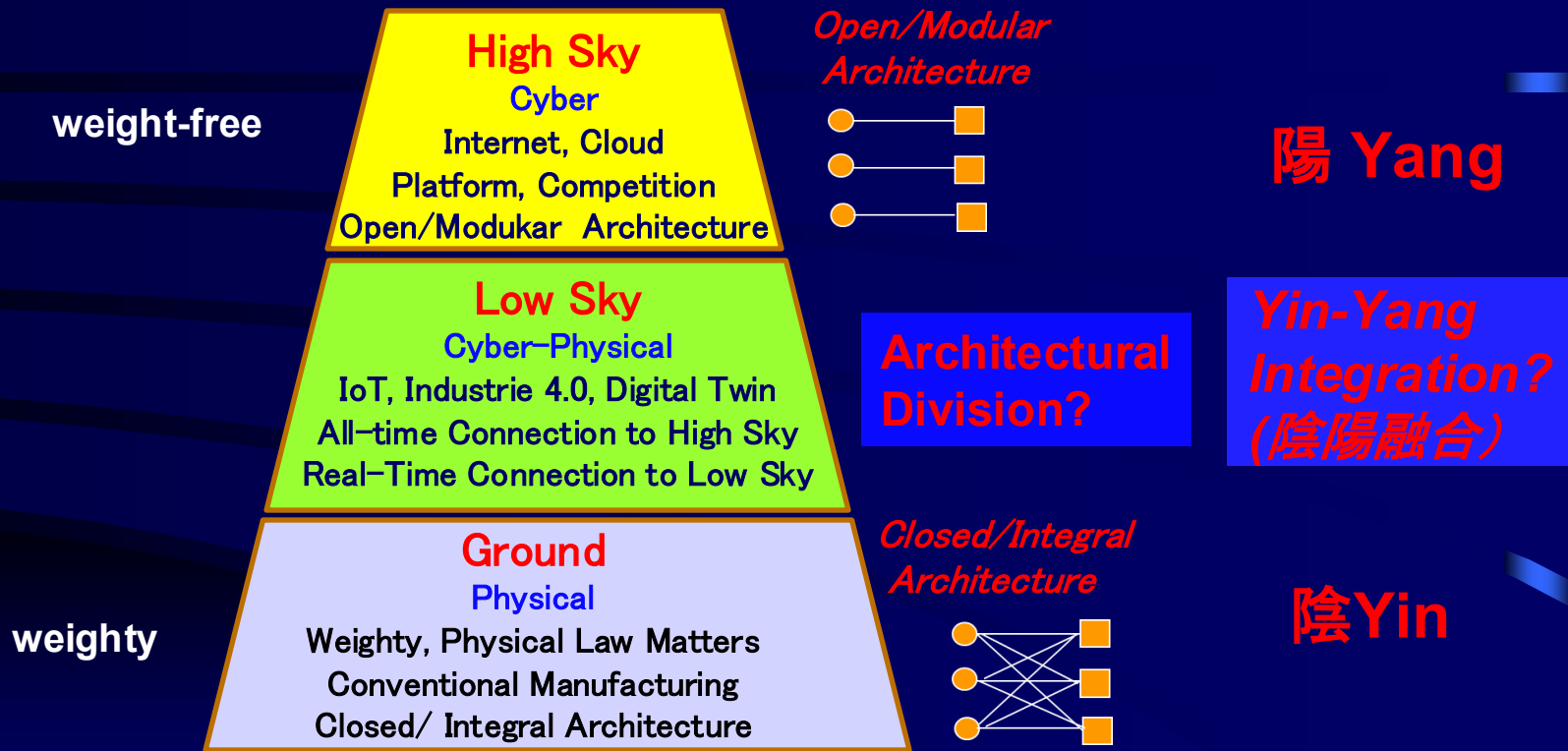
Hypothesis: Software-architecture Division at High Sky, Low Sky and Ground Layer



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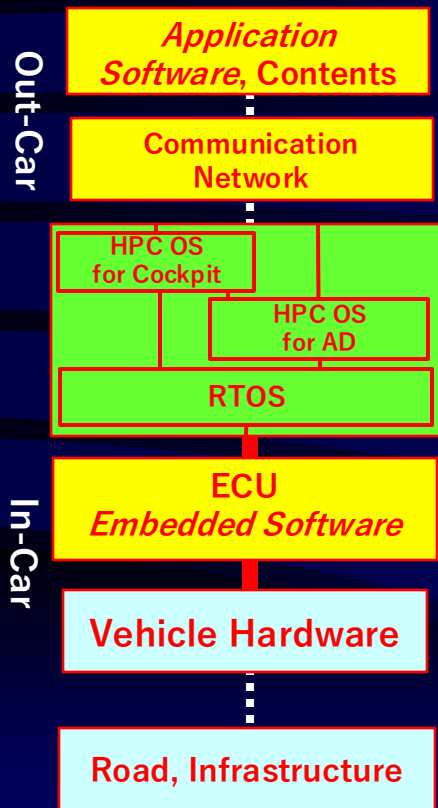


Hypothesis: Software-architecture Division at High Sky, Low Sky and Ground Layer



SDVs as "Software-architecture Divided Vehicles" ?

SDVs' Design Layers



Modular Architecture

Modular
Integral

Divided?

Integral Architecture

Industry Structure

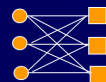
High Sky



Vehicle to Internet
Mobility Service, Infotainment

Google (Android Auto)
Apple (CarPlay; iOS)
SDL (Smart Device Link)
IVI (In-Vehicle Infotainment)

Low Sky



Ground

Vehicle to Vehicle/Infra.

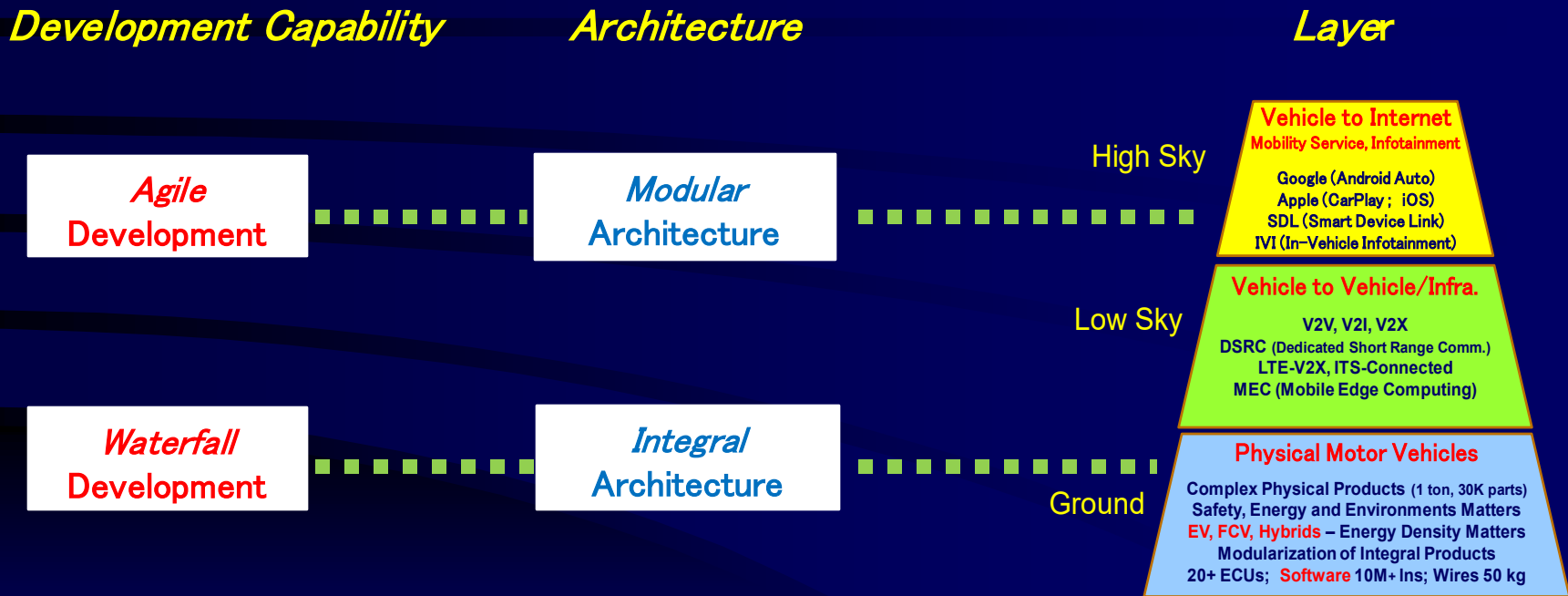
V2V, V2I, V2X
DSRC (Dedicated Short Range Comm.)
LTE-V2X, ITS-Connected
MEC (Mobile Edge Computing)

Physical Motor Vehicles

Complex Physical Products (1 ton, 30K parts)
Safety, Energy and Environments Matters
EV, FCV, Hybrids – Energy Density Matters
Modularization of Integral Products
20+ ECUs; **Software** 10M+ lns; Wires 50 kg

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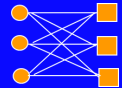
Software Architecture Types and Development Capability



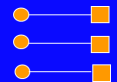
SDVs as Software-architecture Divided Vehicle

When a product is divided into two subsystems of
different architecture types (e.g., integral and modular subsystems),
international cooperation/alliance between two firms of
different capability types may lead to its competitive advantage.

German/Japanese Firms: relatively higher *coordination capability*
Design-based comparative advantage in complex *integral products*
Such as high-functional automobiles' mechanical hardware
and *in-car embedded software* in *ECUs* for vehicle *safety* and driving performance.



Chinese Firms: relatively higher *specialization capability*
Design-based comparative advantage in simpler *open-modular products*,
such as consumer appliance, digital products,
and *out-car application software* for vehicle *infotainment*.



Theoretical Frameworks for International Hardware-Software Alliance

Question: Why Do Incumbent Automakers (Japanese, German, etc.) Choose Overseas R&D in Certain Software Development?

Mainstream Answer (Dunning 1979, Williamson 1975, Rugman 1981, etc.)

1. **Firm-specific Advantage** in High-functional and Integral Hardware
(Design-based comparative advantage in the home country)
2. **Location-specific Advantage** based on **Comparative Advantages**
(Design-based comparative advantage in the local-market country)
3. **Internalization/Externalization** of Design Transactions between R&D Sites
(Internalize, e.g., M&A, if **transaction cost** > coordination cost)

Configuration and Coordination (Porter 1986, 1990)

1. **Configuration** (optimal locations) by “**competitive advantage of nations**”
2. **Coordination** between R&D sites by **M&A, alliance, licensing**, etc.

SDV (Software-defined Vehicle) for Chinese Market

- Product Architecture Matters –

Hypotheses: Design-based Comparative Advantage applied to SDVs

1. Firm-specificity: Automaker X (e.g., Japanese, German), with higher **coordination capability**, tend to develop **integral** (coordination-intensive) subsystem, e.g., **vehicle hardware and embedded software (ECUs)**, at its homebase R&D center (*Homebase-exploiting*: Kuemmerle 1997)
2. Location-specificity: Automakers X may make utilize higher **specialization-capability** of overseas R&D sites (e.g., US, China) for developing **modular** (coordination-saving) components, e.g., **application software for infotainment**, if this local sites have design-based comparative advantage in modular products (*Homebase-augmenting*: Kuemmerle 1997)
3. Internalization/Externalization: Transactions of open-modular products (e.g., software) tend to be less internalized (e.g., alliance vs. M&A)

Three Basic Types of Hardware-Software Links

Three types of hardware-software links that Automaker X may choose

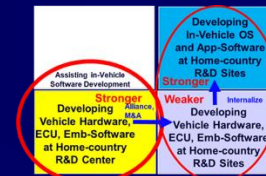
1. Vertical Link: Automaker X (e.g., Japanese, German) develops vehicle hardware, embedded software (ECUs), in-vehicle OS, and key application software internally or with its Tier-1 suppliers in vertically integrated manners.



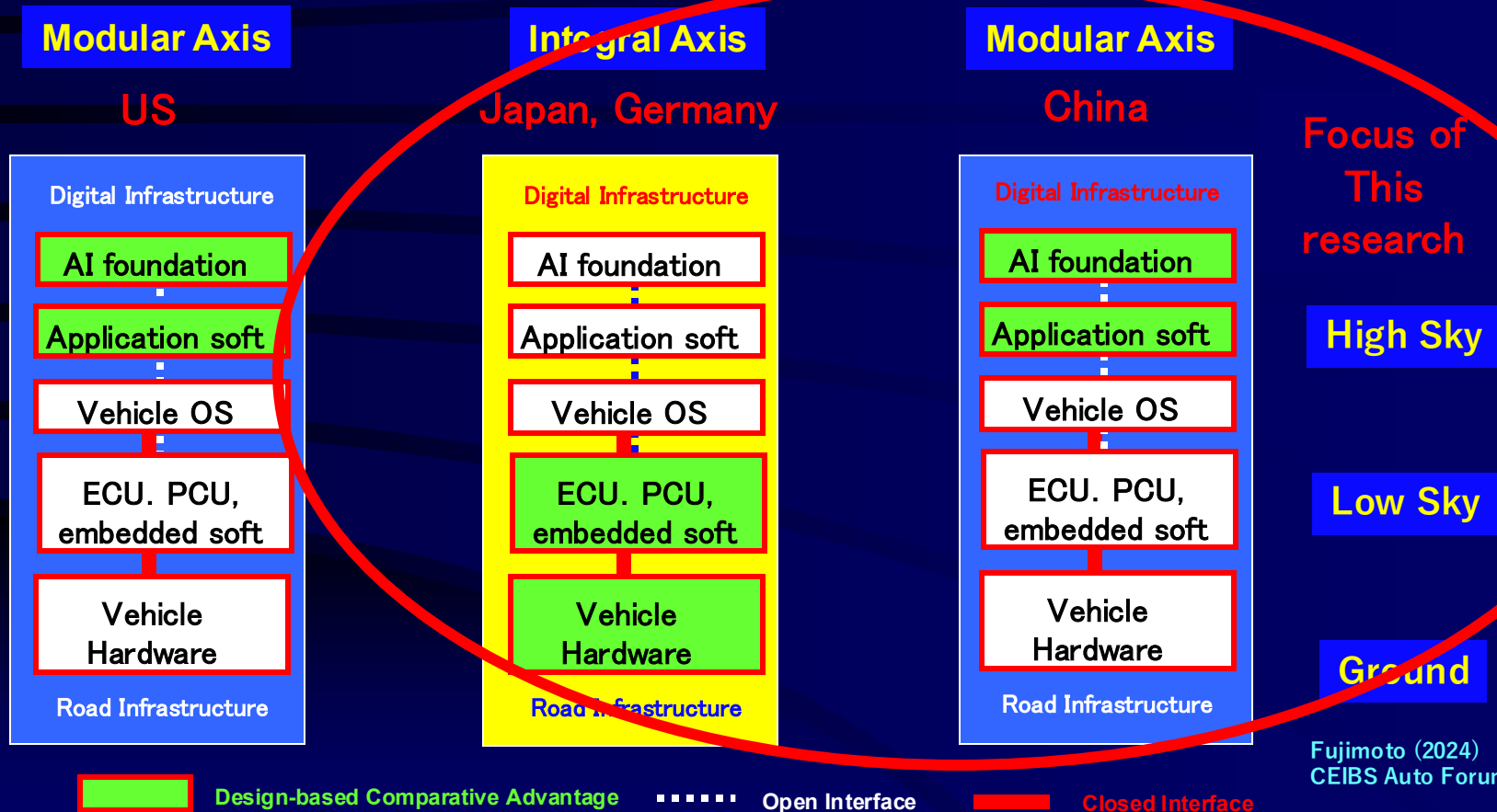
2. Diagonal Link: Automaker X (e.g., Japanese, German), with Tier-1, develops vehicle hardware and embedded software (ECUs). In-vehicle OS may be developed by X, or jointly with local software firms. Application software for infotainment is developed by local forms.



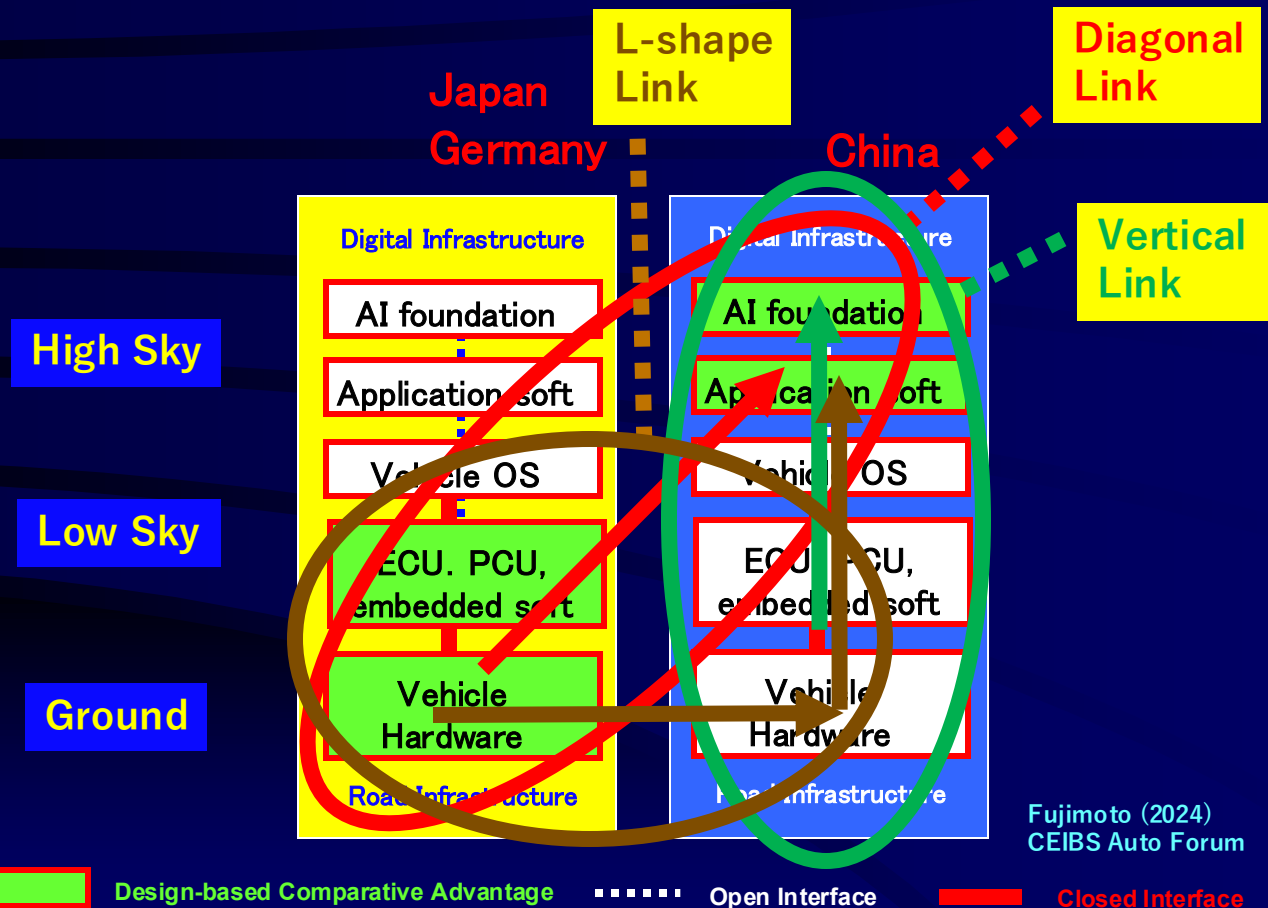
3. L-shape Link: Automaker X, by international alliance or M&A, ties up with local automaker Y and utilize their vertically-integrated software resources.



Hardware-Software Division of Design-based Comparative Advantages in SDVs

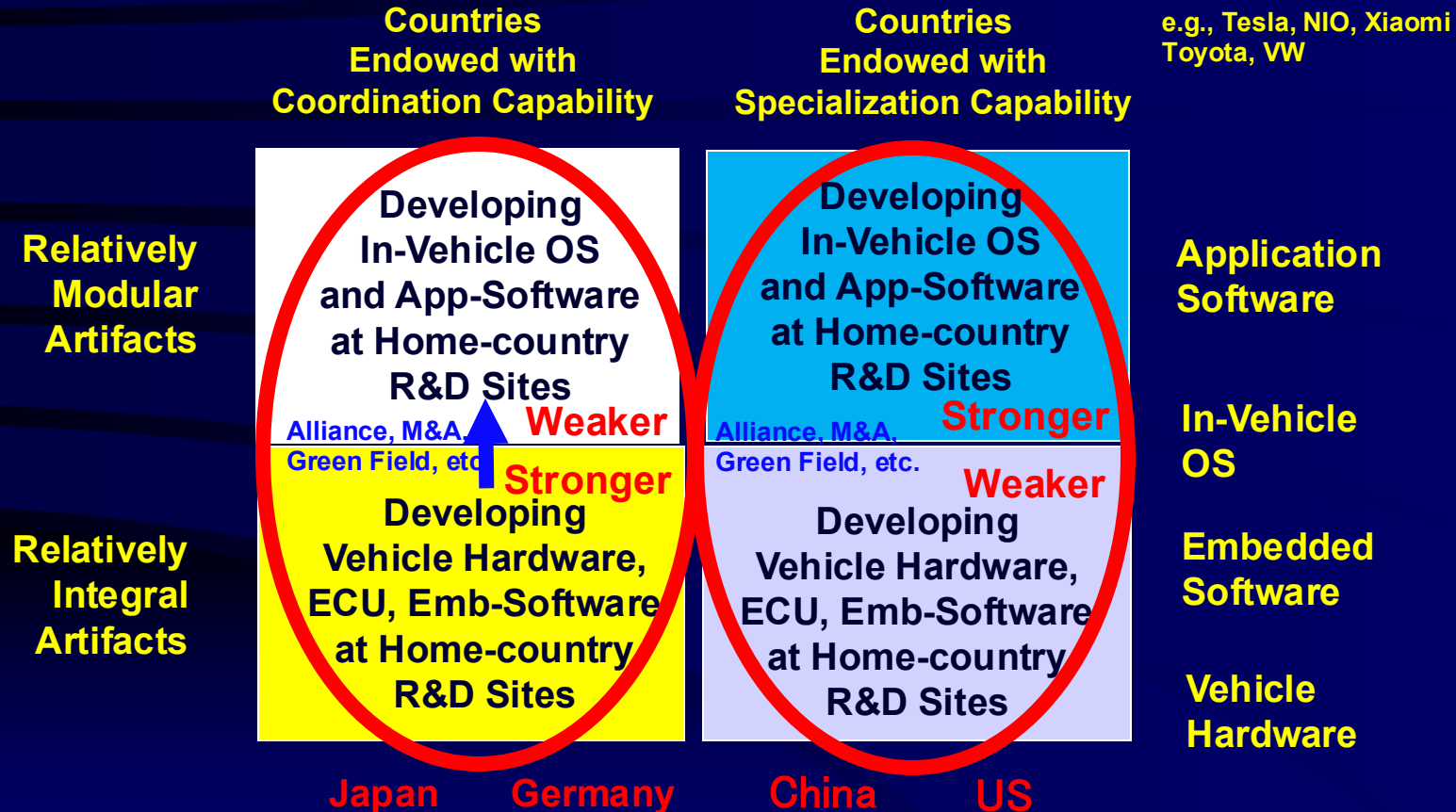


Vertical, Diagonal, L-shape Alliance between High-Sky Firms and Ground Firms

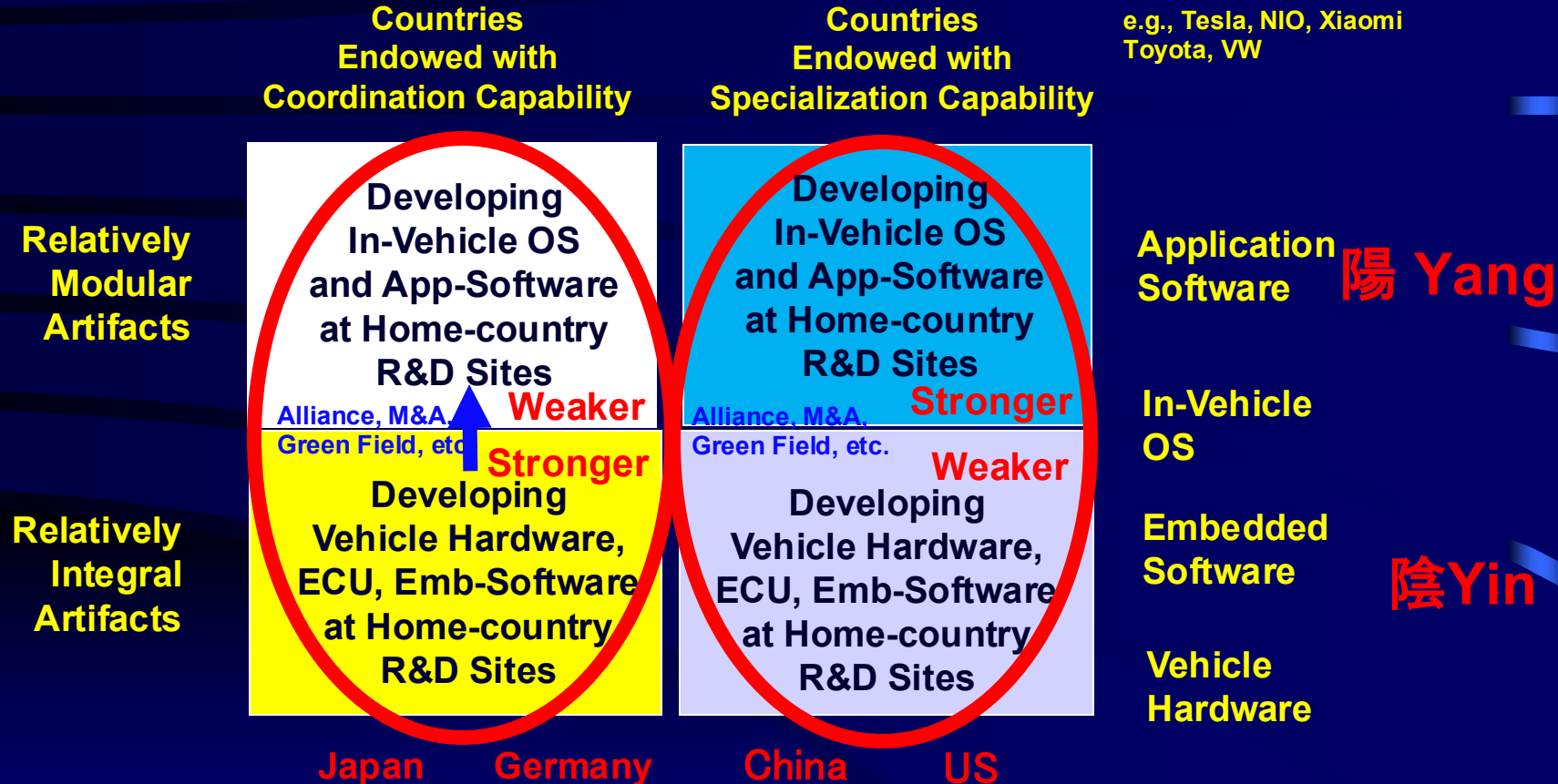


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1. Vertical Links within Home Countries

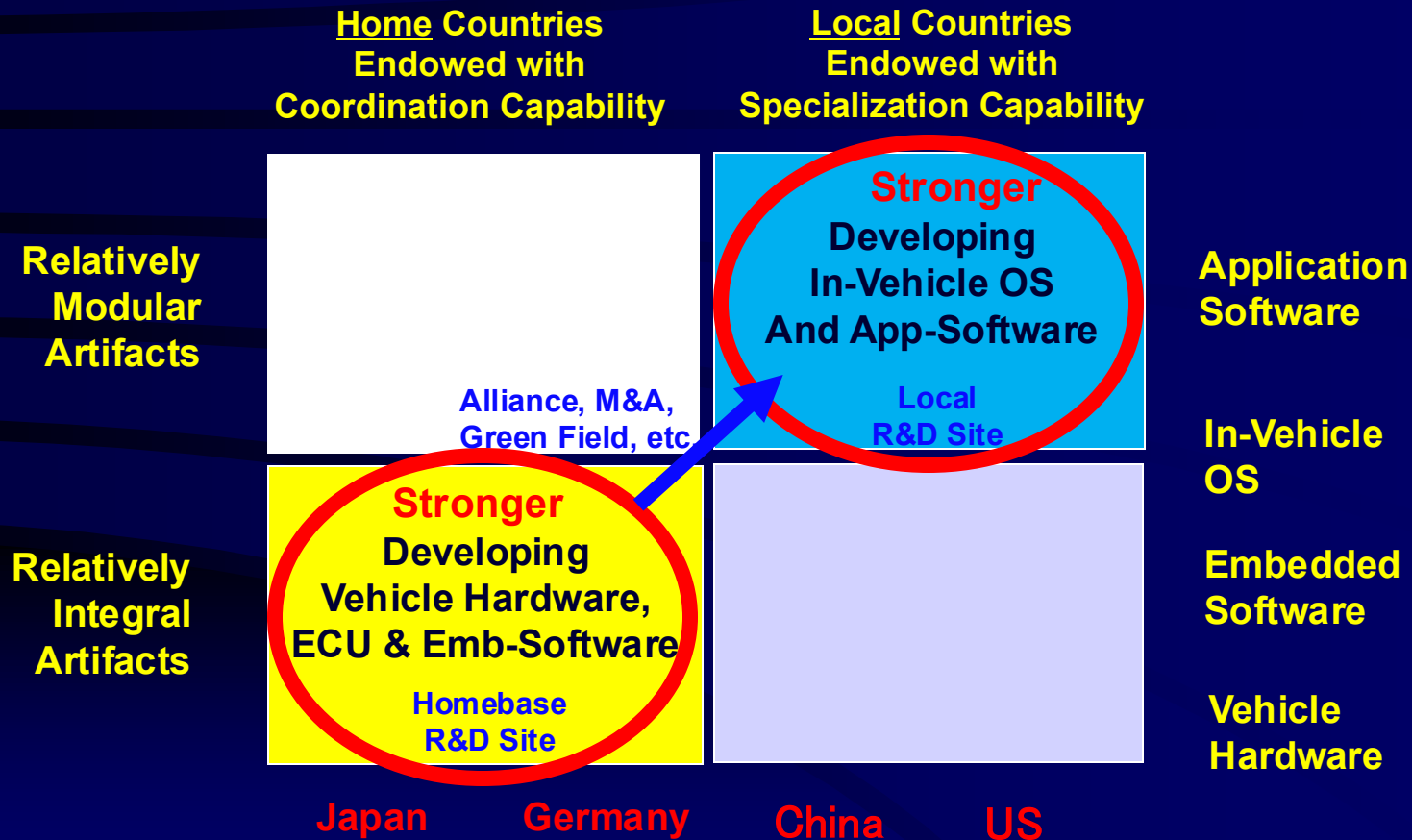


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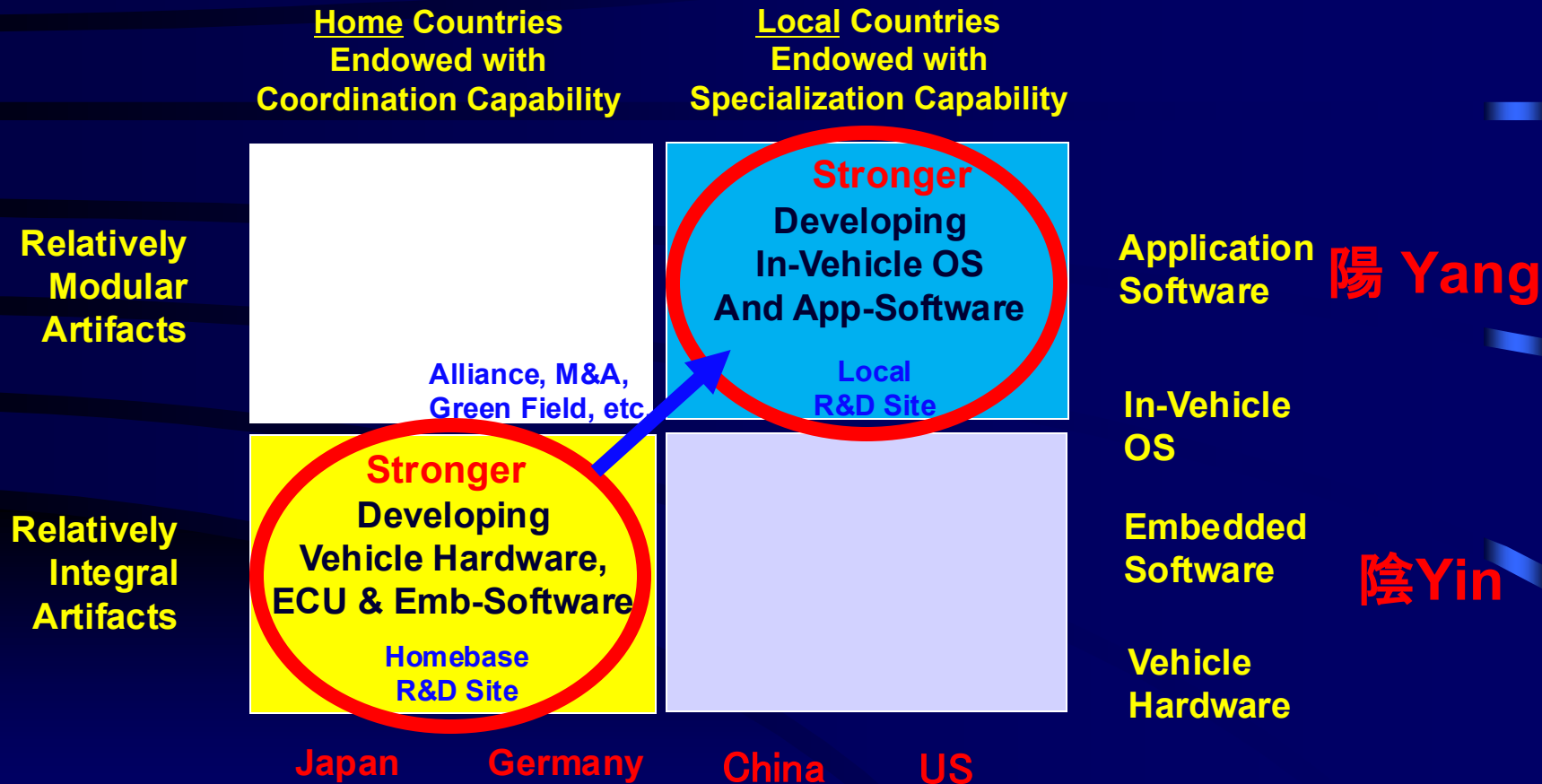


e.g., Tesla, NIO, Xiaomi
Toyota, VW

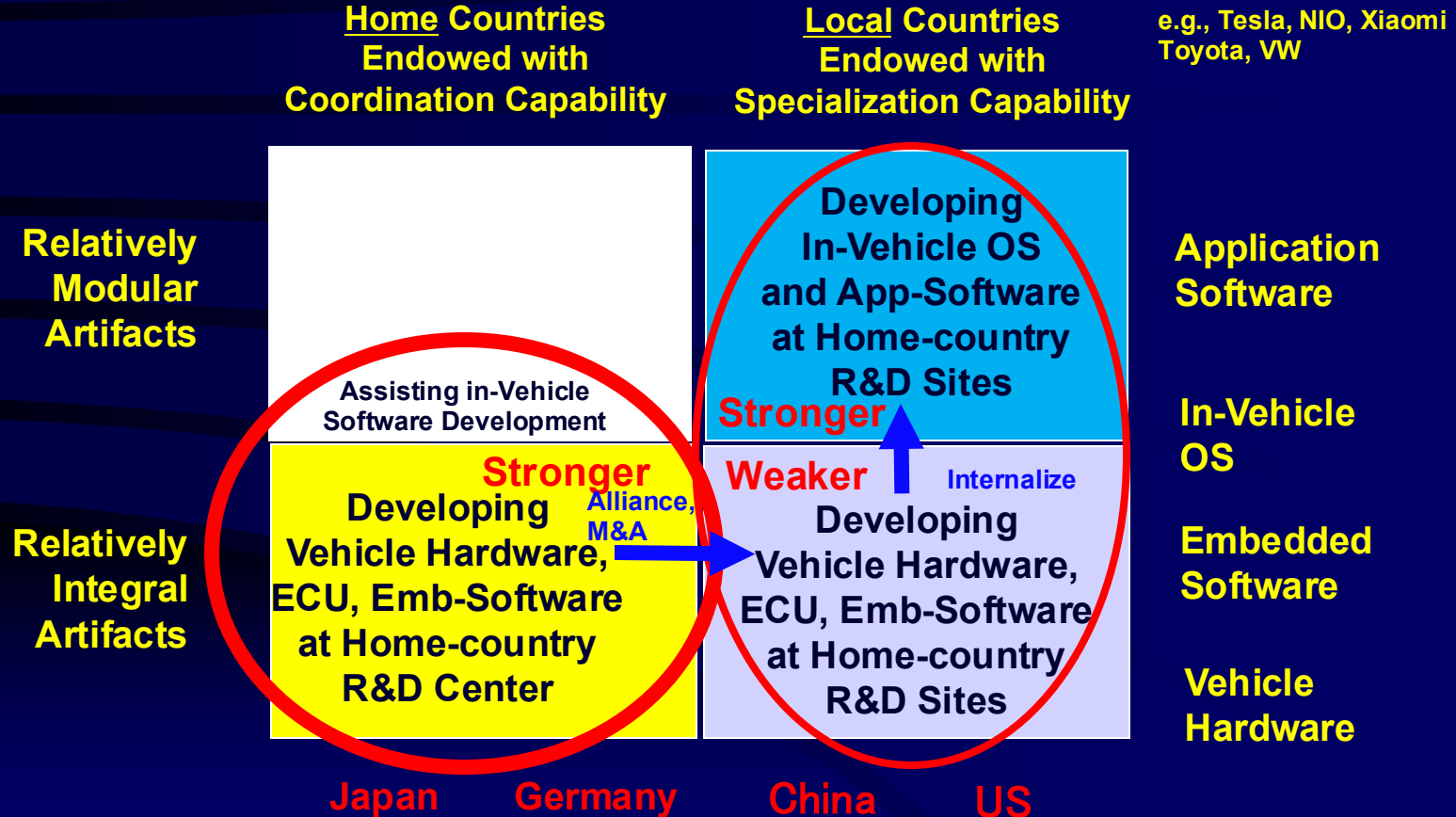
2. Diagonal Links between Stronger Hardware/Software Sites



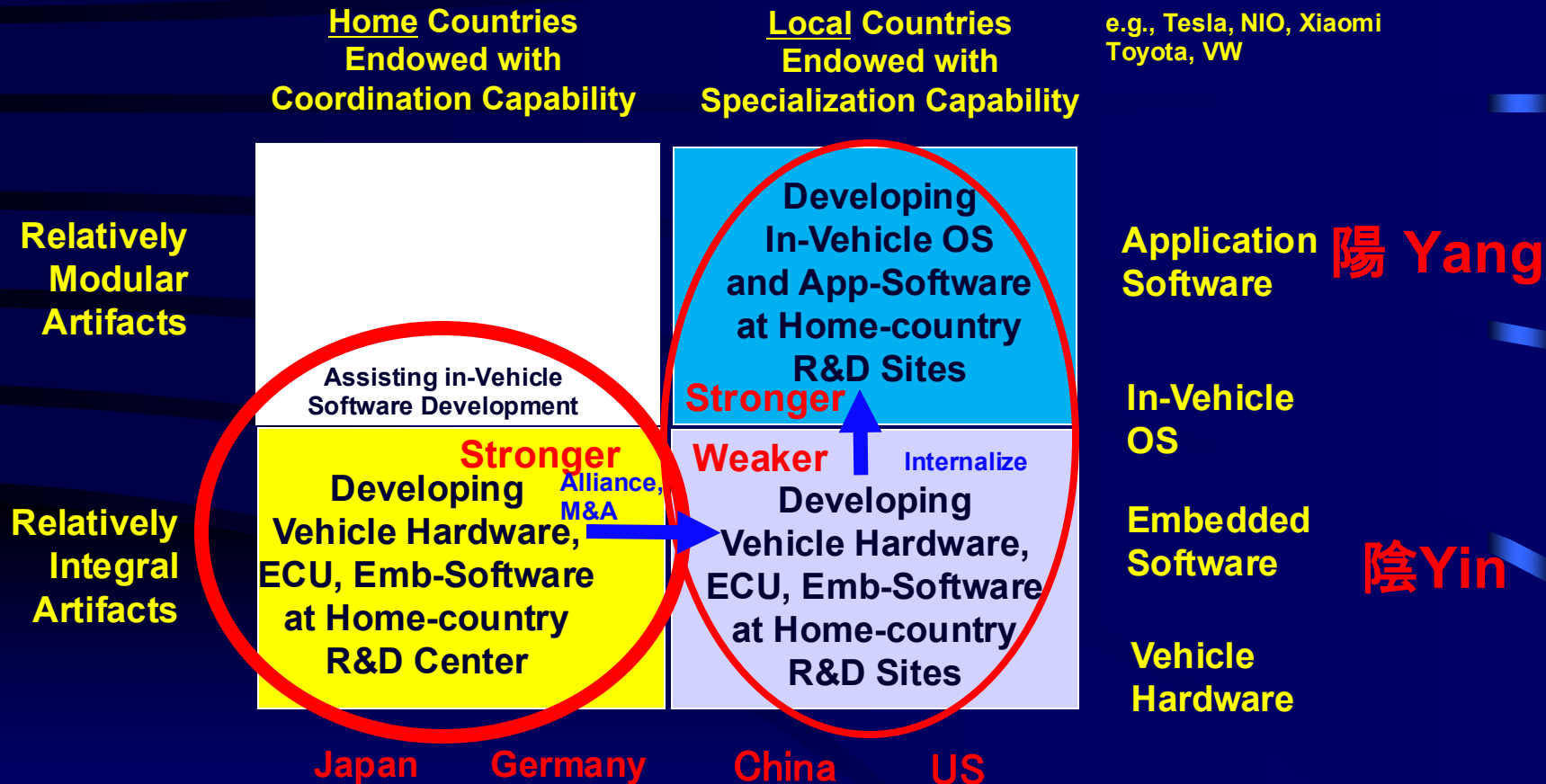
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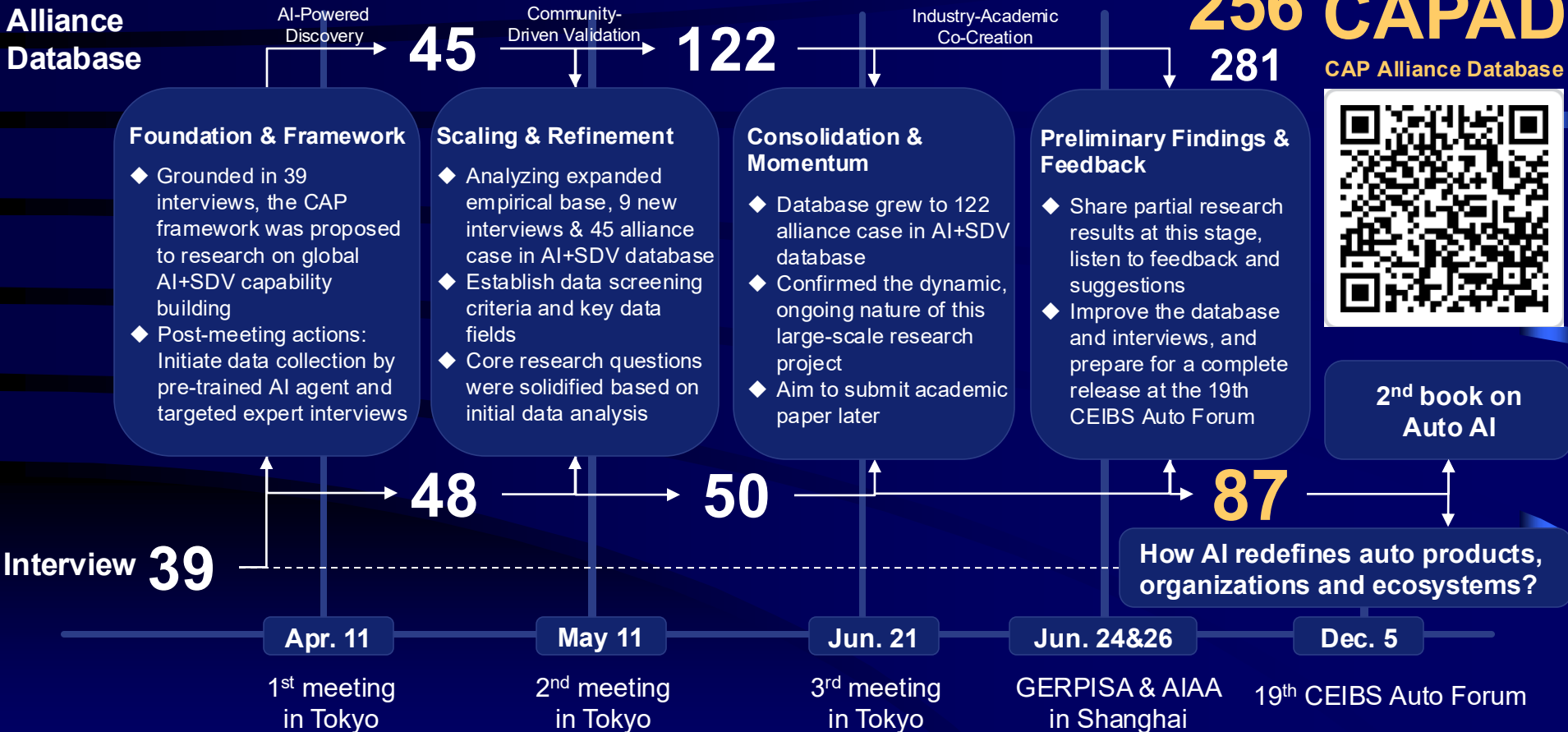
3. L-Shape Links between Hardware- & Software-driven Automakers



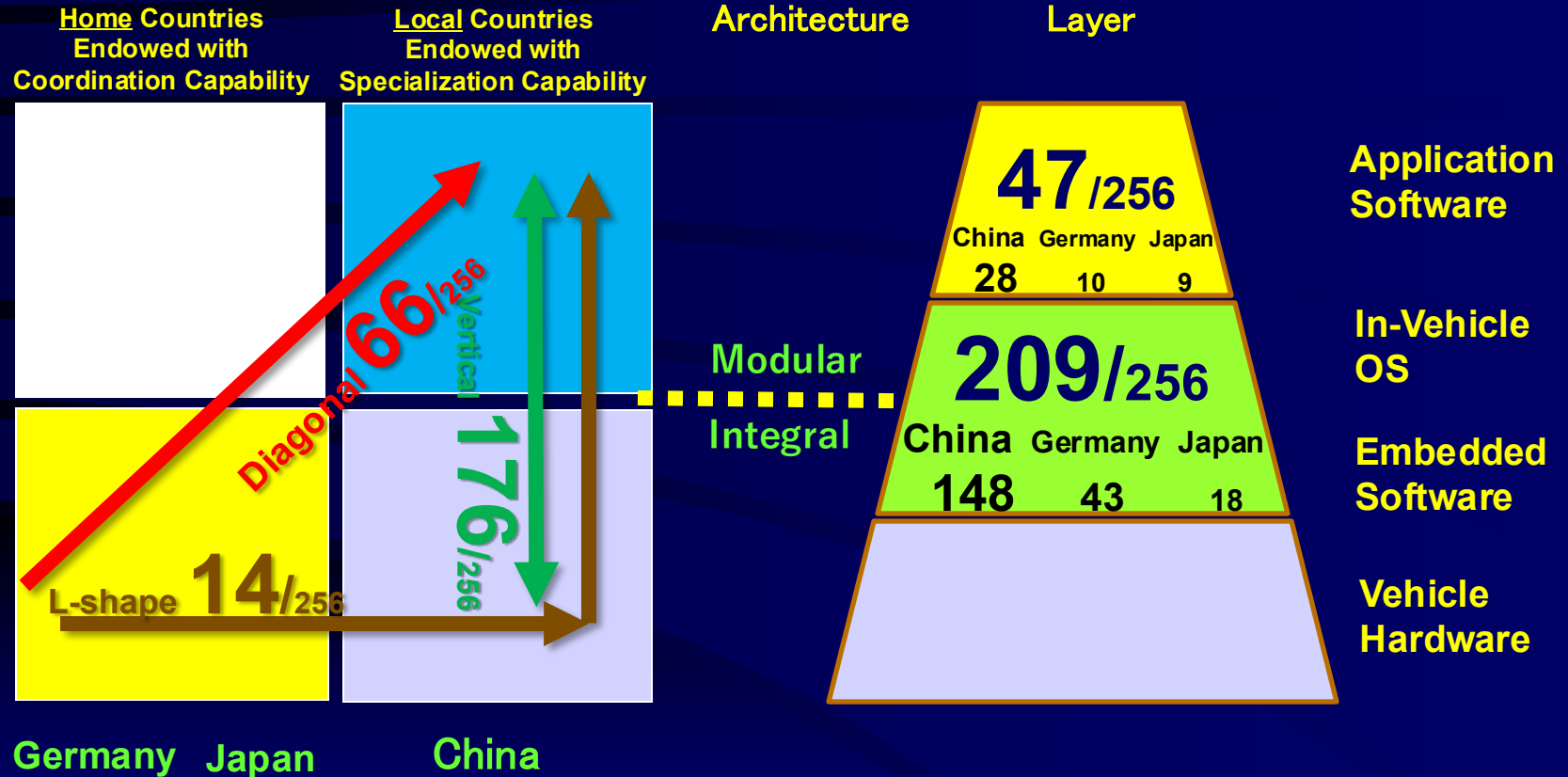
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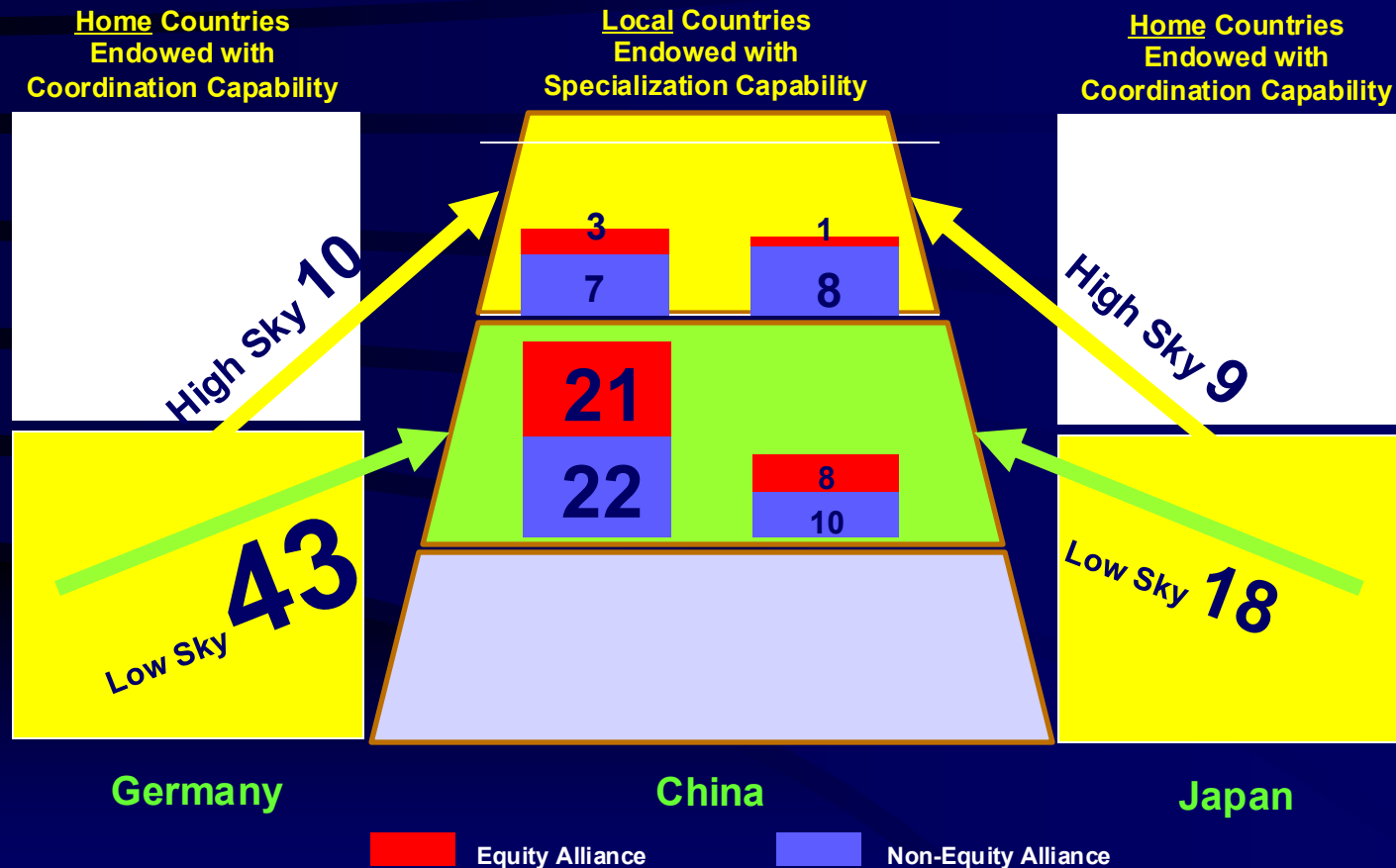
A Living Database as Foundation: The Journey to Enable Co-Creation



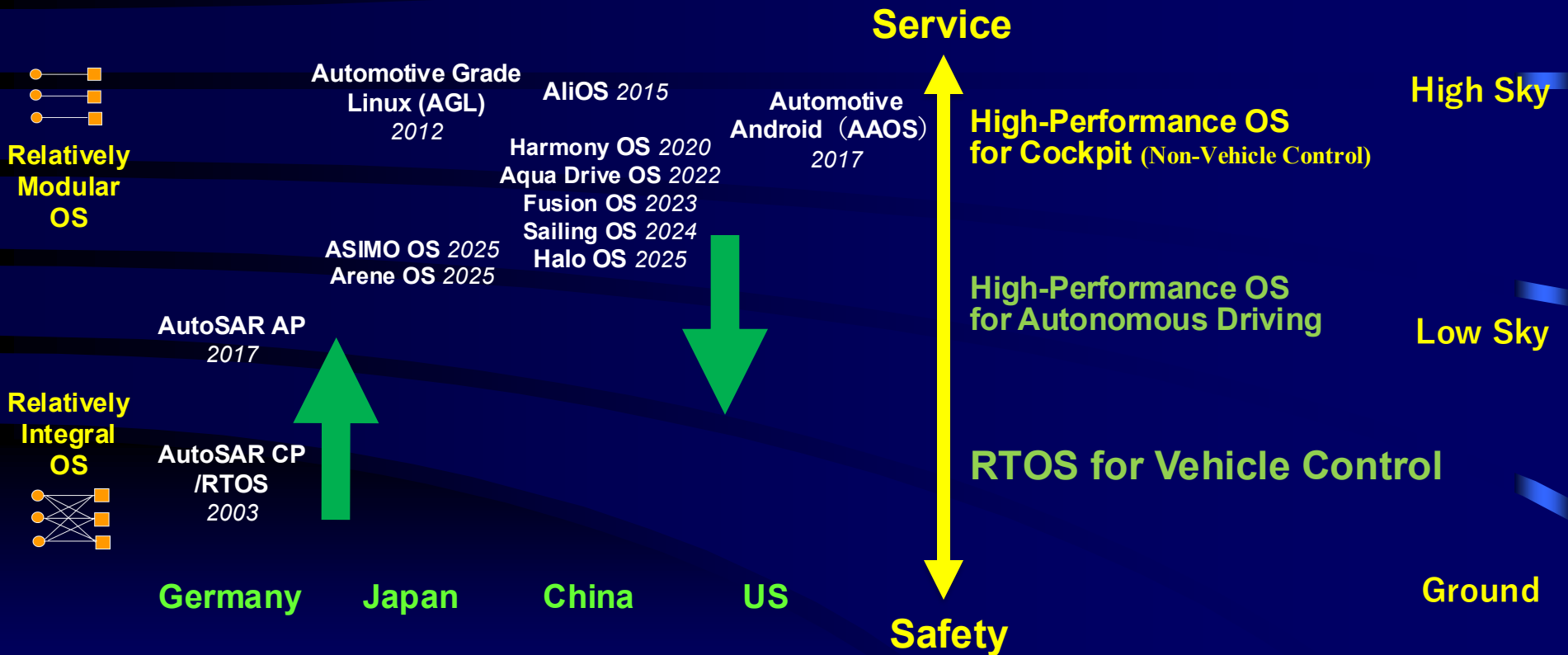
Unveiling the Alliance Landscape: Where Capabilities Fuse



A Deeper Dive: How Architecture Shapes Partnership Models



Integral vs. Modular Architecture in Vehicle OS



Putting Theory to Practice: Volkswagen's Diagonal & L-Shape Strategy

Year	Type of Alliance	Volkswagen Group	AI Digital Software Partner	Joint Venture
2016	Non-equity Alliance	Volkswagen	DiDi	
2017	Equity Alliance - Joint Venture (50%)	Volkswagen	Mobvoi	Volkswagen-Mobvoi
2017	Equity Alliance - Investment	Volkswagen	Mobvoi	
2018	Equity Alliance - Joint Venture (50%)	Volkswagen L-Shape Link FAW		FAW Audi Mobility
2018	Non-equity Alliance	Audi	Huawei	
2018	Non-equity Alliance	Volkswagen	Baidu	
2018	Equity Alliance - Joint Venture (40%)	Volkswagen	DiDi	Juzhong Smart Mobility
2019	Equity Alliance - Joint Venture (16.66%)	Volkswagen L-Shape Link FAW Volkswagen		MOSI
2019	Equity Alliance - Joint Venture (34%)	Volkswagen L-Shape Link FAW Volkswagen	SAIC Volkswagen+GoFun	Yishouche
2022	Equity Alliance - Investment	Volkswagen	Mobvoi	
2022	Equity Alliance - Joint Venture (60%)	Volkswagen's CARIAD	Horizon Robotics	CARIAD Horizon
2023	Equity Alliance - Joint Venture (49%)	Volkswagen's CARIAD	Thundersoft	CARThunder
2023	Equity Alliance - Investment (4.99%)	Volkswagen L-Shape Link Xpeng		
2024	Equity Alliance - Sequential Alliance	Volkswagen L-Shape Link Xpeng		
2024	Equity Alliance - Sequential Alliance	Audi L-Shape Link SAIC		
2025	Equity Alliance - Investment (Angel)	Porsche	Horizon Continental	
2025	Equity Alliance - Sequential Alliance	Volkswagen L-Shape Link Xpeng		

Answer 1: SDV as Software-Divided Vehicle

Q1 Why did German and Japanese automakers, with *design-based comparative advantage* in high-performance automobile hardware, lag behind their US (e.g., Tesla) and Chinese rivals in developing **SDV-EV**?

Hypothesis: **SDV** consists of two architecturally different areas:

- ① relatively *closed-integral* areas, e.g., vehicle *hardware* and *embedded-software* for vehicle/engine control and safety;
- ② relatively *open-modular* areas, e.g., *application software* for infotainment, AI, etc.

German/Japanese automakers tend to have design-based comparative advantage in ①, but NOT in ②

Answer 2: Chinese Automakers have better access to Chinese modular software firms for infotainment

Q2 How could Chinese EV specialists quickly establish its software-hardware integration of SDV-EV?

Hypothesis: *SDV-EV* fits Chinese automobile/software firms capabilities well.

- ① Chinese firms have design-based competitive advantage in *application software* and AI with *open-modular architecture*.
- ② Chinese automakers have *less competitive disadvantage* in mechanically simpler *BEV*, compared with more complex ICE, and significant competitive advantage in automobile *battery*.
- ③ Chinese automakers have *better access to Chinese software firms* for co-developing or adopting application software through purchasing, relational contracts, *vertical alliance*, M&A, etc.

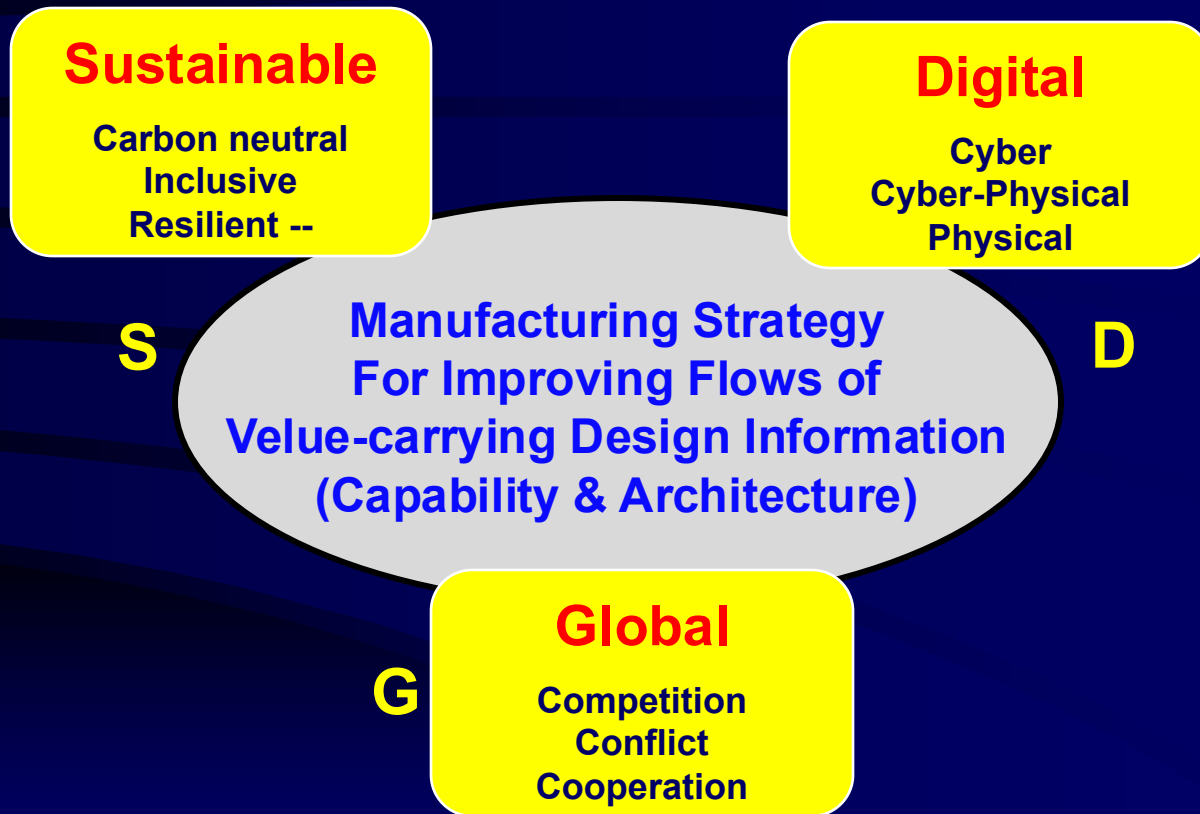
Answer 3: Japanese/German Carmakers utilize diagonal or L-shape links for developing SDV for Chinese market

Q3 How are German and Japanese automakers trying to catch up with the US and Chinese rivals for developing SDV-XV with multiple powertrains (EV, HEV, PHEV, ICE, etc.)?

Hypothesis: *Japanese/German automakers make alliance with Chinese firms*

- ① *Japanese/German firms*, with higher advantage in vehicle hardware and embedded software, try to tie up with *Chinese software/auto firms* with higher advantages in application software and AI, for combining their competitive advantages.
- ② The main way of this international hardware–software ties include *diagonal links* between Japanese/German hardware/firmware and Chinese application software firms. If this is difficult, *L-shape links* between the Japanese/German automakers and Chinese automakers may be chosen,
- ③ The link may *be relational transaction, alliance, M&A*, etc., Depending upon the architectural types of the software (integral/modular, open/closed)

Big SDG (Sustainable, Digital and Global) Challenges



Conclusion

A **Software-Defined Vehicle (SDVs)** consists of two architecturally different zones (i.e., *architecture-divided vehicle*): **closed-integral** (e.g., **hardware** and **embedded software** for safety); **open-modular** (e.g., **application software** for infotainment) layers.

German & Japanese automakers, endowed with **coordination capability**, tend to have design-based competitive advantage in SDV's integral part, including **hardware, embedded software and in-car OS for safety**.

Chinese ICT/software firms, endowed with **specialization capability**, may have design-based competitive advantage in SDV's modular part, including **application software, AI and out-car OS for infotainment**.

Chinese automakers enhanced competitive advantages in **SDV-EV**, where their competitive disadvantages in hardware is less serious.

German & Japanese automakers may accelerate **international inter-firm collaboration** with Chinese ICT/software firms or SDV-EV makers for **combining the design-based competitive advantages** of the two groups.

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